B. Tech. in Mechanical Engineering

Department of Mechanical Engineering

Introduction

Mechanical Engineering is one of the broadest and most influential engineering disciplines, contributing significantly to sectors such as manufacturing, energy, transportation, aerospace, robotics, healthcare, defence, smart infrastructure, and sustainable technologies. With rapid advancements in automation, artificial intelligence, digital manufacturing, and intelligent systems, the role of modern mechanical engineers has evolved from conventional system design to multidisciplinary technology development and innovation. The revised B.Tech. programme in Mechanical Engineering is designed to develop engineers with strong analytical foundations, practical engineering skills, computational proficiency, interdisciplinary exposure, and innovation-oriented thinking. The curriculum integrates classical mechanical engineering principles with emerging technological domains such as Artificial Intelligence (AI), Machine Learning (ML), Smart Automation, Robotics, Industry 5.0, Internet of Things (IoT), Digital Fabrication, Computational Engineering, and Advanced Manufacturing.

The programme adopts a balanced approach involving Strong engineering science fundamentals, Laboratory-intensive experiential learning, Design-oriented engineering education, Computational and simulation-based analysis, Interdisciplinary problem solving, Industry and research exposure, Professional ethics and societal awareness.

The curriculum structure is carefully organized to ensure gradual progression from engineering fundamentals to advanced applications and specialization. The initial semesters focus on mathematics, science, engineering mechanics, computing, workshop practice, and communication skills. Subsequent semesters introduce core mechanical engineering subjects such as thermodynamics, fluid mechanics, mechanics of solids, heat transfer, manufacturing, control systems, and machine design. Advanced semesters provide flexibility through electives in robotics, automation, smart manufacturing, computational methods, thermo-fluids, advanced materials, and autonomous systems, supported by internships, design projects, and research-oriented B.Tech. projects.

The programme also emphasizes hands-on learning through laboratory courses, engineering design modules, digital fabrication practices, industrial exposure, and project-based learning. Students are encouraged to develop innovation and entrepreneurial capabilities while addressing real-world engineering problems related to sustainability, clean energy, intelligent systems, advanced manufacturing, and societal needs.

The programme aligns with national priorities in advanced manufacturing, energy security, automation, defence technologies, sustainability, and digital transformation while preparing graduates for careers in industry, higher education, research organizations, start-ups, and public sector enterprises.

Objective of the Program

The key objectives of the programme are:

- To provide students with strong foundations in mechanical engineering sciences, engineering analysis, and applied mathematics.
- To develop the ability to design, analyze, manufacture, and optimize engineering systems and processes through theoretical, experimental, and computational approaches.

- To integrate emerging technologies such as Artificial Intelligence, Machine Learning, Robotics, Smart Automation, IoT, Industry 5.0, and Digital Manufacturing into mechanical engineering education.
- To promote interdisciplinary thinking and innovation-driven problem solving for addressing industrial, societal, and sustainability challenges.
- To provide extensive laboratory, design, fabrication, and project-based learning experiences for practical engineering competence.
- To inculcate professional ethics, leadership qualities, teamwork, communication skills, and lifelong learning attitude among students.
- To prepare graduates for successful careers in industry, research organizations, entrepreneurship, higher education, and public service.

Expected Graduate Attribute

Graduates of the B.Tech. programme in Mechanical Engineering are expected to possess:

- Strong understanding of core mechanical engineering domains including thermo-fluids, solid mechanics, manufacturing, design, automation, and control systems.
- Ability to apply engineering principles, computational methods, and experimental techniques to solve real-world engineering problems.
- Capability to design and develop mechanical systems, components, products, and manufacturing processes under practical and sustainability constraints.
- Proficiency in the use of modern engineering tools, simulation software, digital technologies, and data-driven approaches.
- Understanding of emerging technologies such as AI, ML, robotics, smart manufacturing, IoT, and autonomous systems in mechanical engineering applications.
- Ability to work effectively in multidisciplinary teams and communicate engineering ideas clearly in oral and written forms.
- Research aptitude, innovation capability, and entrepreneurial mindset for technology development and product innovation.
- Awareness of societal, environmental, ethical, and sustainability considerations in engineering practice.
- Commitment towards professional integrity, lifelong learning, and technological leadership.

Learning Outcome

Upon completion of the programme, graduates will be able to:

- Apply principles of mathematics, basic sciences, and engineering sciences to analyze and solve mechanical engineering problems.
- Identify, formulate, and model engineering problems involving mechanics, thermo-fluids, manufacturing, materials, and dynamic systems.
- Design and conduct experiments, analyze data, and interpret engineering results using scientific and statistical methods.
- Design mechanical components, systems, and processes considering functionality, manufacturability, safety, sustainability, and economic constraints.
- Use modern computational tools, simulation platforms, CAD/CAM software, numerical techniques, and digital technologies for engineering analysis and design.
- Apply concepts of AI, ML, robotics, automation, smart manufacturing, IoT, and Industry 4.0 in mechanical engineering applications.

- Develop hands-on engineering skills through laboratory work, fabrication practices, internships, design activities, and capstone projects.
- Function effectively as an individual and as a member or leader in multidisciplinary teams.
- Communicate technical concepts, engineering analyses, and project outcomes effectively through reports, presentations, and discussions.
- Understand professional ethics, environmental responsibilities, and societal impacts associated with engineering decisions.
- Engage in independent learning, research, innovation, and professional development to adapt to evolving technological challenges.

Table 1: Course Categories, Credit Distribution and Credit Structure of B.Tech. Programme

S.N.	Course Type	Course Category	Credits	Minimum Total
1	Institute Core (I)	Engineering (IE)	16	39 (27.6%)
		Science (IS)	17	
		HSS (IH)	06	
2	Programme Core (P)	Programme Compulsory (PC)	66	90 (63.8%)
		Programme Electives (PE)	18	
		(B.Tech.) Project (PP)	6	
3	Open (O)	Electives (OE)	12	12 (8.5%)
Total (Graded)				141 (100%)
4	Essential Audit (Non-taught)	Humanities (NH)	3	12
		Engineering (NE)	3	
		Industry-Academia Summer Internship outside IIT Jodhpur (two summers, minimum 45 days)	4	
		Design Credit	2	
Total for award of Degree (Graded + Non-Graded)				153

Table 2: Semester-wise plan for B.Tech Mechanical Engineering

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	I Semester						**II Semester**				
IE	Engineering Mechanics	3-0-0	3	-	3	IE	Introduction to Machine Learning	3-0-2	5	-	4
IE	Introduction to Computer Science	3-0-2	5	-	4	IS	Chemistry	3-0-0	3	-	3
IS	Physics	3-0-0	3	-	3	IS	Introduction to Bioengineering	2-0-2	4	-	3
IH	HSS I	3-0-0	3		3	IS	Chemistry Lab	0-0-2	2	-	1
IS	Physics Lab	0-0-2	2	-	1						
IS	Mathematics I	2-1-0	3	-	3	IS	Mathematics II	2-1-0	3	-	3
IE	Engineering Drawing	0-0-2	2	-	1	IE	Engineering Workshop	0-0-2	1	-	1
NE	Engineering Design I	0-0-2	2	1	-	NE	Engineering Design II	0-0-2	2	1	-
NH	Communication Skill	0-0-2	2	1	-	IH	HSS II	3-0-0	3		3
NH	Social Connect and responsibilities	0-0-1	1	0.5	-	NE	Introduction to Profession	0-0-2	2	1	
NH	Performing Arts /Sports	0-0-1	1	0.5	-						-
Total					**18**	**Total**					**18**
	III Semester						**IV Semester**				
PC	Fluid Mechanics	3-1-0	4		4	PC	Heat and Mass Transfer	3-0-0	3	-	3
PC	Kinematics and Dynamics of Machines	3-0-0	3	-	3	PC	Mechanics of Solids	3-1-0	4	-	4
PC	Thermodynamics	3-1-0	4	-	4	PC	Engineering Materials and Processing Techniques	3-0-0	3		3
PC	Casting Welding Forming	3-0-0	3	-	3	PC	Mechanical Measurements and Metrology	3-0-0	3		3
PC	Mechanics and Manufacturing Lab	0-0-6	6	-	3	PC	Mechanical Measurements and Characterization Lab	0-0-8	8	-	4
IE	Environment and Sustainability	3-0-0	3	-	3						
Industry-Academia Summer Internship outside IIT Jodhpur										**2**	**-**
Total					**20**	**Total**				**2**	**17**
	V Semester						**VI Semester**				
PC	Design of Machine Elements	3-1-0	4		4	PC	Introduction to Smart Automation	3-0-0	3		3
PC	Machining Science	3-0-0	3		3	PC	Production & Operations Management	3-0-0	3		3

PC	Power Systems Engineering	3-0-0	3		3	PC	Control of Mechanical Systems	3-0-0	3		3
PC	Computer-assisted Fabrication	3-0-0	3		3	PC	Applied Thermo-Fluids	3-0-0	3		3
PC	Design, Manufacturing, and Thermal Power Lab	0-0-6	6	-	3	PC	Automation, Control and Conditioning Lab	0-0-8	8	-	4
PE	Programme Elective	3-0-0	3	-	3	PE	Program Elective	3-0-0	3		3
NH	Professional Ethics I	1-0-0	1	1	-						
Industry-Academia Summer Internship outside IIT Jodhpur										**2**	**-**
Total					**19**	**Total**				**4**	**19**
	VII Semester						**VIII Semester**				
PP	B.Tech Project 1	0-0-3			3	PP	B.Tech Project 2	0-0-3			3
PE	Programme Electives	2X (3-0-0)	6		6	PE	Programme Electives	2X (3-0-0)	6		6
OE	Open Electives	2X (3-0-0)	6		6	OE	Open Electives	2X (3-0-0)	6		6
Total					**15**	**Total**					**15**
Total of Graded and Non-Graded Credit										**6**	**141**
Non-Graded Design Credits and Industry-Academia Summer Internship outside IIT Jodhpur										**6**	**-**
Grand Total										**12**	**153**

Table 3: Programme Compulsory Courses

Course	*LTP*	*CH*	*GC*	*GC*
Fluid Mechanics	3-1-0	4		4
Mechanics of Solids	3-1-0	4	-	4
Thermodynamics	3-1-0	4	-	4
Casting Welding Forming	3-0-0	3	-	3
Heat and Mass Transfer	3-0-0	3	-	3
Design of Machine Elements	3-1-0	4	-	4
Engineering Materials and Processing Techniques	3-0-0	3		3
Mechanical Measurements and Metrology	3-0-0	3		3
Kinematics and Dynamics of Machines	3-0-0	3		3
Machining Science	3-0-0	3		3
Power Systems Engineering	3-0-0	3		3
Computer-assisted Fabrication	3-0-0	3		3
Introduction to Smart Automation	3-0-0	3		3
Production & Operations Management	3-0-0	3		3
Control of Mechanical Systems	3-0-0	3		3
Applied Thermo-Fluids	3-0-0	3		3
Mechanics and Manufacturing Lab	0-0-6	6	-	3
Mechanical Measurements and Characterization Lab	0-0-8	8	-	4
Design, Manufacturing, and Thermal Power Lab	0-0-6	6	-	3
Automation, Control and Conditioning Lab	0-0-8	9	-	4
Total				**66**

Table 4:Area-wise Programme Elective Courses

Course	*LTP*	*CH*	*GC*	*Course Level*
Solid Mechanics and Design				
Basics of Finite Element Analysis	3-0-0	3	3	400
Mechanical Vibrations	3-0-0	3	3	400
Continuum Mechanics	3-0-0	3	3	700
Engineering Fracture Mechanics	3-0-0	3	3	700
Mechanics of Smart Materials and Structures	3-0-0	3	3	700
Structural Health Monitoring	3-0-0	3	3	700
Robotics and Automation				
Introduction to Mechatronics	3-0-0	3	3	400
Human-centered automation and Industry 5.0	3-0-0	3	3	700
Introduction to Robotics	3-0-0	3	3	400
Planning and Decision Making for Robots	3-0-0	3	3	700
Mobile Robots	3-0-0	3	3	700
Vision for Automation	3-0-0	3	3	700
Manufacturing				
Micro and nano Manufacturing	3-0-0	3	3	400
Welding Science and Technology	3-0-0	3	3	400
Fundamentals of Metal Forming	3-0-0	3	3	700
Advanced Machining	3-0-0	3	3	700
Casting and Allied Processes	3-0-0	3	3	400
Computer-Aided Manufacturing	3-0-0	3	3	700
Thermo-Fluids				
Computational Fluid Dynamics and Heat Transfer	3-0-0	3	3	700
Conduction and Radiation	3-0-0	3	3	700
Convective Heat Transfer	3-0-0	3	3	700
Numerical Methods for Mechanical Engineers	3-0-0	3	3	400
Combustion Technology	3-0-0	3	3	700
Micro nano Flows	3-0-0	3	3	400
Computer-Aided Inspection and Quality Control	3-0-0	3	3	700
Systematic Embodiment Design	3-0-0	3	3	700
Mechatronics System Design	3-0-0	3	3	700
Sports Biomechanics	3-0-0	3	3	700
AI, ML, and Big Data for Mechanics and Design 'or' Artificial Intelligence for Manufacturing Applications 'or' Machine Learning for Thermofluids	3-0-0	3	3	700

Table 5: About Lab Courses

Mechanics and Manufacturing Lab	Kinematics and Dynamics of Machines Lab Manufacturing Processes Lab Fluid Mechanics Lab
Mechanical Measurements and Characterization Lab	Heat and Mass Transfer Lab Experimental Solid Mechanics Lab Mechanical Measurements and Metrology Lab Engineering Materials and Processing Techniques Lab
Design, Manufacturing, and Thermal Power Lab	Design of Machine Elements Lab Power Systems Engineering Lab

	Machining and Automated Manufacturing Lab
Automation, Control and Conditioning Lab	Smart Automation Lab Control of Mechanical Systems Lab Design and Computational Mechanics Lab Thermo-Fluids Engineering Lab

Table 6: Minor in ME (Compulsory Courses)

Cat	Course	LTP	GC
SC	Mechanics of Solids*	3-1-0	4
SC	Kinematics and Dynamics of Machines	3-0-0	3
SC	Casting Welding Forming	3-0-0	3
SC	Machining Science	3-0-0	3
SC	Thermodynamics**	3-1-0	4
SC	Fluid Mechanics	3-1-0	4
Total			**21**

*Design of machine elements, if Mechanics of solids is already PC for the programme

** Heat and Mass Transfer***, if Thermodynamics is already PC for the programme.

***Power Systems Engineering, if Thermodynamics and Heat and Mass Transfer both are already PC for the programme

Title	**Engineering Mechanics**	Number	MEL1XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1st Year (All branches)	Type	IE
Prerequisite	Nil		

Objectives

1. To apply knowledge of statics and dynamics to particles and rigid bodies

2. To develop the ability to solve numerical problems based on statics and dynamics principles.

Learning Outcomes

1. To model practical structural problems using concepts of free body diagrams and equilibrium conditions

2. To solve dynamic problems using Newton's Law, Work-Energy and Impulse-Momentum Principles

Contents

Statics [17 Lectures]: Basic dimensions in Mechanics, Law of dimensional homogeneity, Vector and Scalar Quantities, Elements of vector algebra. Moment of a force about a point/axis, Couple, Free Body Diagram, Equations of Equilibrium, Static indeterminacy, Trusses and Beams, Shear force and bending moment diagrams, Coulomb Friction, Surface contact friction, Transmission of power through belt. Pappus-Guldinus Theorems, second moments and product of Area, Principal axes.

Dynamics of Particles [10 Lectures]: Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion. Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles.

Dynamics of Rigid bodies [12 Lectures]: Translation/Rotation of rigid bodies, time derivative of vector for different references. Parallel axis theorems, Rotational Pure rotation of a body of revolution about its axis of revolution/combined with translation. Three dimensional rotation, moment of inertia tensor, relation between angular momentum and torque in three dimensions, Motion in non-inertial frames, centrifugal force, Coriolis force/acceleration, rate of change of vector in inertial and rotating frames.

Reference Books

1. Shames, I., (2003), *Engineering Mechanics*, Prentice Hall

2. Gross, D., Hauger, W. and Schrder J., (2012), *Engineering Mechanics*, Springer

3. Meriam, J. L., Kraige, L. G., (2002), *Engineering Mechanics*, John Wiley & Sons

4. Harbola, M K., (2013), Engineering Mechanics, CENGAGE Learning

Self Learning Material

1. Harbola, M K., *Engin*eering Mechanics, NPTEL Course Material, Department of Physics, Indian Institute of Technology Kanpur, https://nptel.ac.in/courses/115104094/

2. Lewin, W. H. G, Physics I: Classical Mechanics - Fall 1999, MIT, http://videolectures.net/mit801f99_physics_classical_mechanics/

Title	**Engineering Drawing**	Number	MEP1XX0
Department	Mechanical Engineering	L-T-P [C]	0-0-2 [1]
Offered for	B.Tech. 1st Year (All branches)	Type	IE
Prerequisite	--		

Objective: Aims to provide a basic understanding of the fundamentals of engineering drawing, standards and conventions of drawing, and the use of engineering drawings in engineering applications.

Learning Outcomes: Students will be able to:
1. Understand and apply engineering drawing conventions and practices.
2. Visualize and represent 2D and 3D engineering components accurately.
3. Interpret engineering drawings used in real-world applications.
4. Develop drafting skills using instruments and/or CAD tools.

Contents:

Introduction [1 Lab]: introduction and importance of engineering drawing; drawing techniques; conventions- ISO and BIS, layout of drawing sheets; Lettering and dimensioning; scales.
Geometrical constructions [1 Labs]: Bisecting a line, drawing perpendicular and parallel lines, dividing a line, and circle. Bisect and trisect an angle. tangents to lines and arcs, construct equilateral triangles, squares, regular pentagons, and regular polygons inscribed in circles.
Orthographic projects [2 Lab]: Theories of projection, multi-view (orthographic) projection; VP; HP; Front view; Top view; Projection of objects placed in all Four Quadrants, First and third angle projections.
Projection of points and straight lines [2 Labs]: Points in 1st, 2nd, 3rd, and 4th quadrants; Line parallel to both the planes, Line perpendicular to one plane and parallel to the other; Line inclined to one plane and parallel to the other plane, Lines inclined to both planes. True lengths and inclinations, Traces of lines.
Projection of planes [2 Labs]: Projections of planes parallel to one of the reference planes, Projections of planes inclined to one reference plane and perpendicular to the other, Projections of oblique planes.
Projection of solids [1 Lab]: Projections of solids in simple positions.
Development of surfaces [2 Labs]: Method of development, Development of surfaces of oblique solids.
Machine Drawing [1 Lab]: bolt and nut assembly.

Text/Reference books

1. Bhatt, N. D., 2002, Elementary Engineering Drawing, Charoter Publishing.
2. Jolhe, D.A., 2008. Engineering Drawing with an Introduction to AutoCAD.
3. Luzzader, W. J. and Duff, J. M., 2008, Fundamentals of Engineering Drawing, Prentice Hall
4. Bethune, J. D., 2007, Engineering Graphics with Autocad, Prentice Hall

Online course material
1. https://nptel.ac.in/courses/112103019
2. http://www.digimat.in/nptel/courses/video/112104172/L01.html

Title	**Engineering Workshop**	Number	MEP1XX0
Department	Mechanical Engineering	L-T-P [C]	0-0-2 [1]
Offered for	B.Tech. 1st Year (All branches)	Type	IE
Prerequisite	--		

Objective: To provide a practical exposure to the different vocational trades within the basic workshop practice framework.

Learning Outcomes: Students will be able to:
1. Demonstration of basic skill set in workshop practices.
2. Development of basic knowledge of hand tools and machine tools in different areas of manufacturing.

Contents:

1.General Safety
Workshop safety rules and personal protective equipment (PPE).
Introduction to Workshop tools and equipment
2.Fitting Shop
Techniques: Filing, hacksawing, drilling, tapping, and reaming.
Projects: V-fit, semi-circular fit, square joint.
3.Carpentry Shop
Techniques: Marking, planning, sawing, chiseling, and turning.
Projects: Half-lap joint, Mortise and Tenon joint.
4.Welding Shop
Techniques: Arc welding equipment, gas welding, brazing, and soldering.
Projects: Butt joint, Lap joint.
5.Machine Shop
Techniques: operations on Lathe, shaper, drilling machine, milling machine
Projects: turning, facing, step-turned pulley
6.Sheet Metal
Techniques: Shearing, bending, riveting, and soldering.
Projects: Tapered tray, conical funnel.
7.Foundry & Forging
Techniques: Sand casting, pattern making, forging tools, and operations.
8.Modern Manufacturing/Digital Fabrication
Introduction to 3D printing.
9. Modern Manufacturing/Digital Fabrication
Introduction to CNC machines.
10. Heat treatment of steels
Introduction and demonstration of heat treatment processes

Laboratory Work:
Woodworking (Pattern making exercise), Preparation of aluminum casting, Machining exercise (turning operations), Welding exercise (Preparation of square butt joints, T-joints using arc welding), Sheet metal fabrication (Preparation of tray, funnel, etc.), Fitting exercise, heat treatment of steels, Demonstration on CNC Lathe, CNC Milling. Demonstration on Rapid Prototyping Technique.

Text Books:

1. Chapman W A J, Workshop Technology- Part 1, Edward Arnold Ltd (2001)
2. Chapman W A J, Workshop Technology- Part 2, Edward Arnold Ltd (2007)

Title	**Casting Welding Forming**	Number	MEL2XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective: The objective of the course is:
1. Impart fundamental understanding of various casting, welding and forming processes.
2. Discuss design methodology and process parameters involved in obtaining defect free components.
3. Discuss tools, techniques and equipment involved in casting, welding and forming processes.

Learning Outcomes: The students are expected to have the ability to:
1. Select appropriate primary manufacturing process and related parameters for obtaining initial shape and size of components.
2. Design and develop adequate tooling linked with casting, welding and forming operations.

Contents:

Metal Casting Processes [13 Lecture]: Introduction to Casting [3 Lectures]: Casting Terminology, Types of pattern and allowances, Applications and Challenges. Design of various parts [5 Lecture]: Riser and Gating system design, Design principles for mould cavity. Solidification [2 Lecture]: Metal flow and solidification; solidification and cooling of casting. Types of casting process and defects [3 Lecture]: Expendable and Permanent Mould Casting Processes, Die casting and centrifugal casting, Casting defects and inspection of castings.

Welding [13 Lecture]: Introduction to Joining Processes [3 Lecture]: Basic requirements and classification of welding processes; Arc Welding Processes, Arc welding principle, Welding power source and its characteristics. Arc Welding Processes [4 Lecture]: Gas Tungsten Arc Welding (GTAW) and Gas Metal Arc Welding (GMAW) processes and their parameters; Submerged arc welding (SAW), and Shielded metal arc welding (SMAW); Advance welding processes [2 Lecture]: Resistance welding processes; Weldability and thermal aspects [3 Lecture]: Concept of weldability of materials; Thermal Effects in Welding (Distortion, shrinkage and residual stresses in welded structures); Welding defects and remedies. Allied processes [1 Lecture]: Soldering and Brazing.

Metal Forming Processes [13 lecture]: Introduction of metal forming process [4 Lecture]: Mechanical behavior of metals in elastic and plastic deformation, stress-strain relationships, Yield criteria, Application to tensile testing, train rate and temperature in metal working; Hot deformation, Cold working and annealing. Metal Working Processes [5 lecture]: Fundamentals of metal working, Analysis of bulk forming processes like forging, rolling, extrusion, wire drawing by slab method, Other sheet metal processes [4 Lecture]: Sheet metal forming processes (Die and punch assembly, Blanking, piercing, bending etc., Compound and Progressive die).

Text Books:
1. Ghosh, A. and Mallik, A. K., (2017), Manufacturing Science, East-West Press.
2. Parmar R. S., (2007), Welding Processes and Technology, Khanna Publishers.

Reference Books:
1. Heine, R. W., Loper, C. R., and Rosenthal, P. C., (2008), Principles of Metal Casting, Tata McGraw Hill.
2. Conne, L. P., (1989), Welding Handbook- Welding Technology Part-1, American Welding Society.
3. Black, S. C., Chiles, V, Lissaman, A. J., and Martin, S. J., (1996), Principles of Engineering Manufacture, Butterworth-Heinemann.
4. Groover, M. P., (2011), Introduction to Manufacturing Processes, Wiley.
5. Kalpakjian, S., and Schmid, S. R., (2001), Manufacturing Engineering and Technology, Pearson.
6. Riggs, J. L., (1976), Production System-Planning Analysis and Control, John Wiley.

Self-Learning Material:
1. D. B. Karunakar, Metal Casting, NPTEL Course Department of Mechanical and Industrial Engineering, IIT Roorkee, Link: https://nptel.ac.in/courses/112107083/
2. P. Biswas, Fundamental of Welding Science and Technology, Department of Mechanical Engineering, IIT Guwahati, https://nptel.ac.in/courses/112107083/
3. P. K. Jha, Principles of Metal Forming Technology, Department of Mechanical and Industrial Engineering, IIT Roorkee, https://nptel.ac.in/courses/112107250/

Title	**Mechanics of Solids**	Number	MEL2XX0
Department	Mechanical Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective:

1. To inculcate fundamental concepts of stress, strain, and deformation of solids
2. Fundamentals of applying equilibrium, compatibility, and forced deformation relationships to structural elements

Learning Outcomes: The students are expected to have the ability to:

1. Apply the fundamentals to analyze structural members and gain physical insight into their stress and strain distributions.
2. Ability to design structural members using dimensions, force-displacement relationships, boundary conditions, loading, and allowable stress

Contents:

Concept of stress [7 Lectures]: Surface and body forces, concept of stress at a point, stress equilibrium, plane stress case: transformation of stresses at a point, principal stresses, and Mohr's circle, St. Venant's principles.
Concept of strain [5 Lectures]: Displacement field, strain at a point, compatibility relations, volumetric strain, plane strain case: transformation of strain at a point, principal strains and Mohr's circle, Strain Rosette;
Material behaviour and mathematical model [4 Lectures]: Concepts of elasticity, response of real materials, Generalized Hooke's law (without and with thermal strains) for isotropic materials, orthotropic, and anisotropic materials.
Application of Elasticity:
One-dimensional axial deformation [2 Lectures]: axial force diagram, basic relations for axial deformations, Stress concentration.
Torsion of circular members [3 Lectures]: Basic relations for torsion of circular members, stress distribution of torsion members.
Pressure Vessels [3 Lectures]: Thin-walled pressure vessel theory, thin-walled spheres, thin-walled cylinders.
Beams [7 Lectures]: Shear force and moment diagrams; moment-curvature relationship for pure bending of beams with symmetric cross-section; bending stress, shear stress; cases of combined stresses; deflection due to bending.
Buckling [3 Lectures]: Concept of elastic instability, Introduction to column buckling, Euler's formula.
Energy methods [5 Lectures]: Concept of strain energy; strain energy for simple structural elements; Castigliano's theorems; principle of minimum potential energy.

Text/Reference books

1. Crandall, S.H., Dahl, N.C., and Lardner, T. J., (1978), An Introduction to the Mechanics of Solids, McGraw Hill, Second Ed. with SI Units
2. Popov, E.P., (1990), Engineering Mechanics of Solids, First Ed. Prentice Hall
3. Beer, F.P, Johnston, E.R. and De Wolf, J.T., (2004), Mechanics of Materials, Tata McGrawHill.
4. Meriam, J.L. and Kraige, L.G., (1980), Engineering Mechanics, Vol. 1: Statics, 2nd Ed. John Wiley

Online course material

Prof. Ajeet Kumar, NPTEL Course, Solid Mechanics, IIT Delhi https://nptel.ac.in/courses/112102284
Prof. Priyanka Ghosh, NPTEL Course, Mechanics of Solids, IIT Kanpur https://nptel.ac.in/courses/105104160

Title	**Fluid Mechanics**	Number	MEL2XX0
Department	Mechanical Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech.	Type	PC
Prerequisite			

Objective:

1. To inculcate the fundamental laws governing fluid flow
2. Help the students to understand the importance and application of fluid mechanics
3. To exemplify concepts of similarity and model testing

Learning Outcomes: The students are expected to have the ability to:

1. Apply fundamental knowledge in modelling and analysis of fluid flow problems in engineering
2. Interpret data from experiments of fluid flows and solve using differential equations and charts

Contents:

Introduction (2 Lectures): Definition and properties, continuum approach, types of fluids, basics of vector calculus and index notations

Fluid statics (5 Lectures): Basic equations, Manometry, Hydrostatic force on submerged bodies, rigid body motion

Integral analysis of fluid flow (5 Lectures): Reynolds transport theorem, conservation equations

Differential analysis (6 Lectures): Kinematics of fluids, deformations in fluid particle, strain rate, vorticity, stream function, potential function, streamlines, pathlines, streaklines, derivation of Navier-Stokes equations

Incompressible inviscid flow (5 Lectures): Euler's equation, Bernoulli's equation, elementary potential flows, superposition of elementary flows, vorticity and vortex flows

Internal Incompressible viscous flow (3 Lectures): Hagen-Poiseuille and Couette flows, skin friction and loss coefficients, venturimeter, orifice meter and Pitot tube

External incompressible viscous flow (5 Lectures): Boundary layer over flat plate, boundary layer thickness, Prandtl-boundary layer equations, momentum integral equation, adverse pressure gradient & separation, drag coefficients

Dimensional analysis and similarity (2 Lectures): Buckingham Pi theorem and examples

Introduction to turbulent flows (3 Lectures): Decomposition of instantaneous velocities, time-averaged continuity and Navier-Stokes equations, Reynolds stresses, turbulent viscosity

Basics of compressible flow (3 Lectures): Acoustic waves, isentropic equations, concept of shock waves

Reference books

1. Fox, R. W., Pitchard, P. J and McDonald, A. T., (2010), Introduction to Fluid Mechanics, 7th Edition, John Wiley & Sons Inc.
2. Kundu, P. K, Cohen, I. M and Dowling, D. R, (2014), Fluid Mechanics, 6th Edn., Associated Press
3. White, F. M., (2003), Fluid Mechanics, 5th Edition, McGraw-Hill

Online course material

1. S. Chakraborty, Introduction to Fluid Mechanics, NPTEL Course Material, Mechanical Engineering, IIT Kharagpur, https://nptel.ac.in/courses/112/105/112105269/
2. V.Shankar, Fluid Mechanics, NPTEL Course Material, Chemical Engineering, IIT Kanpur, https://nptel.ac.in/courses/103/104/103104044/#
3. Nishith Verma, Fluid Mechanics, NPTEL Course Material, Chemical Engineering, IIT Kanpur, https://nptel.ac.in/courses/103/104/103104043/

Title	**Thermodynamics**	Number	MEL2XX0
Department	Mechanical Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective: The objectives of the course are:

1. To understand thermodynamics laws and various thermodynamic processes
2. To comprehend work producing and work consuming cycles
3. To inculcate the importance of change of state responsible for physical and chemical changes.
4. To introduce broader aspects of thermodynamics across disciplines and applicability of laws of thermodynamics and entropy principle in the areas ranging from design of prime movers to design of algorithms.

Learning Outcomes: The students are expected to have the ability to:

1. Effectively apply laws of thermodynamics in the practice of engineering.
2. Assess and quantify thermodynamic efficiencies using the second law of thermodynamics.
3. Conduct comprehensive and rigorous treatment of classical thermodynamics while retaining an engineering perspective
4.

Contents:

Basic concepts [2 Lectures]: Thermodynamic system, Properties, State, Process, Thermodynamic equilibrium, Pressure measurement, Zeroth law, Temperature measurement.

Concept of energy and energy transfer[7 Lectures]: Work and Heat, Internal energy, First Law of Thermodynamics, Work transfer and its value for different processes, Heat transfer, state functions, Equation of state for ideal and real gases, Gas mixtures; Differential form of First Law, First Law and various Processes.

Thermodynamic properties of pure substance[4 Lectures]: Phase change process of pure substance and their property diagram, Use of steam table and Mollier diagram.

Energy analysis of closed and open systems [8 Lectures]: First law analysis for closed systems, Steady flow energy equation, Flow processes, Throttling process, Joule-Thomson effect, Energy analysis of open systems, and Analysis of First law for unsteady flow.

Second law of thermodynamics [4 Lectures]: Reversible and Irreversible processes, Kelvin-Planck and Clausius statement, Heat engines, Heat pumps, Carnot cycle.

Entropy [8 Lectures]: Definitions of entropy, statistical definition of pressure, temperature, entropy, Change of entropy for various processes, machine learning and coding application examples, Equilibrium, Spontaneity relations for Entropy, Helmholtz free energy, Gibbs free energy.

Thermodynamic property relations [3 Lectures]: Maxwell relations, Clausius-Clapeyron equation

Exergy [3 Lectures]: Definition of exergy, second law efficiency, Irreversibility, Exergy balance.

Text/Reference books:

1. Moran, M. J. & Shapiro, H. N., (2011), Fundamentals of Engineering Thermodynamics, 7th Edition, Wiley.
2. Reynolds, W. C., & Colonna, P. (2018). Thermodynamics: Fundamentals and engineering applications. Cambridge University Press.
3. Cengel, Y. A. & Boles, M. A., (2010), Thermodynamics: An Engineering Approach, McGraw Hill Education
4. Sonntag, R. E. & Wylen, V., (1998), Fundamentals of Thermodynamics, Wiley
5. Zemansky M., (2017), Heat and Thermodynamics, McGrawhill

Online course material:

Prof. Keith A. Nelson & Prof. Moungi Bawendi., Thermodynamics and Chemical Kinetics, MIT OpenCourseWare, Link: https://ocw.mit.edu/courses/chemistry/5-60-thermodynamics-kinetics-spring-2008/

Title	**Design of Machine Elements**	Number	MEL3XX0
Department	Mechanical Engineering	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective: Understand the nomenclature and selection of machine elements and be able to design machine components taking into consideration stress concentration, failure, and fatigue.

Learning Outcomes: The students are expected to have the ability to:

1. Understand the different failure modes of machine components and their role in engineering design.
2. Compare the design options involving various machine elements used in the machine design process.

Contents:

Fundamentals

Introduction to Engineering Design [5 Lectures]: Mechanical engineering design, Design considerations, Design methods, Engineering materials and their mechanical properties, types of codes & standards in design, Selection of preferred sizes, Behavior of ductile & brittle material, Stress-strain diagram for various materials. Material selection in design.

Design for strength [5 Lectures]: Stresses in members subjected to axial, shear, Bending, Torsional & Eccentric loading, Failure criteria such as Maximum Normal and Shear Stress theory, distortion energy theory; Factor of safety; Stress concentration factor; Selection and design of Keys, Cotter & Knuckle Joints, and couplings.

Fatigue [7 Lectures]: Fatigue failure, Endurance limit & strength, Fatigue strengths for reversed bending and reversed axial loading, torsional loading, biaxial loading, Effect of mean stress on fatigue strength; Fatigue life prediction with randomly varying load; Problems on the design of members for finite & infinite life in members subjected to individual & combined loading. Fatigue crack growth.

Applications

Design of shaft [5 Lectures]: Types, Design of solid & hollow shaft on strength and rigidity basis with steady loading subjected to pure torsion. Design of shafts carrying pulleys & gears (Combined loading). ASME Code for shaft design; Classification and selection of bearings.

Design of joints [5 Lectures]: Design of riveted, bolted, and welded Joints.

Design of Helical Springs [3 Lectures]: Torsion bar spring, Coil spring stress and deflection equations, stress and strength analysis for helical compression springs; Buckling analysis; Design of helical compression spring for static and fatigue loading.

Clutches and Brakes [3 Lectures]: Disk clutch, Disk brake, energy absorption and cooling, Cone clutches and brakes;

Flexible Mechanical Elements [3 Lectures]: Flat belts, V-belts, Toothed belts; Roller chains.

Gear Design [2 Lectures]: Spur gear, nomenclature, force analysis. Beam strength of gear tooth.

Text/Reference books

1. Ashby M. F., (2011), Materials Selection in Engineering Design, Butterworth-Heinemann.
2. Shigley, J. E., Mischke, C., Nisbett, K., and Richard, B., (2008), Mechanical Engineering Design, McGraw Hill Education
3. Spotts, M. F. & Shamp, T. E., (2003), Design of Machine Elements, Prentice Hall
4. Juvinall, R.C. and Marshek, K.M., 2020. Fundamentals of machine component design. John Wiley & Sons
5. Mahadevan, K., Reddy, K.B. (2013), Design Data Handbook for Mechanical Engineers, CBS Publishers & Distributors

Online course material

Prof. G. Chakraborty, Prof. S.K. Roychowdhury, Prof. B. Maiti , NPTEL Course, Design of Machine Elements I, IIT Kharagpur https://nptel.ac.in/courses/112105124

Title	**Engineering Materials and Processing Techniques**	Number	MEL2XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective: The objective of the course is

- The knowledge of engineering materials from metals to polymers and ceramics and about their processing

Learning Outcomes: The students are expected to have the ability to:

- Have an understanding about different classes of materials.
- Develop the knowledge of fundamental principles related to materials structure, properties and Processing.

Contents:

Basic Introduction: History of materials, Atomic structure. Atomic bonding in solids (**Lecture 3**)

Structures of Materials: Crystal structure and Bravias lattice, Unit cell and Miller indices, Crystal system, coordination number and packing fraction. Crystallographic directions and planes. Crystalline materials: single crystals, polycrystalline materials. Non–crystalline materials, Imperfection in solid (point defects, line defects, surface defects and volume defects) (**8 Lectures**)

Engineering Materials: Classification of engineering materials: ferrous alloys and nonferrous alloys, ceramics (glasses, glass- ceramics, refractories, abrasives, traditional ceramics, advanced ceramics), polymers, composites. Salient features of metals, ceramics, glass, polymers, and composites (**Lecture 3**)

Classification of Manufacturing Processes: The processes-shaping, joining, finishing. Process- Property Trajectories, Principles of solidification: Nucleation & growth kinetics, homogeneous & heterogeneous nucleation, solidification of metal in ingot mould, directional solidification, dendrite growth, types of segregation, porosity, coring and homogenization. (**7 Lectures**)

Concepts of alloy system and explanation of terms like system, component, phase, degree of freedom, constituents of an alloy, phase rule, lever rule, Hume-Rothery rules, Equilibrium diagrams and their classification based on solubility of components in liquid and solid states, cooling curves, concept of super cooling, morphology and distribution of phases, effect of non- equilibrium cooling on morphology. Binary phase diagrams **(6 lectures)**

Heat treatment: Purpose of heat treatments, equilibrium and non-equilibrium cooling, nucleation, grain growth and kinetics, TTT and CCT diagrams, common heat treatments like annealing, normalizing, hardening and tempering, hardenability: Jominy end-quench test, hardenability curves, martempering and austempering, surface hardening (carburizing, nitriding, flame and induction hardening) (**6 Lectures**).

Fe-Fe3C phase diagrams: Allotropic changes, construction of iron cementite and iron- graphite phase diagrams, critical temperatures and invariant reactions, microstructure and properties of different alloys (both steels and cast irons) (**6 lectures**).

Text/Reference books

1. Introduction to Materials Science, William D. Callister Jr., David G. Rethwisch, John Wiley & Sons; 8th Edition, 2010
2. Introduction to Materials Science, V. Raghvan, Prentice Hall India Learning Private Limited, New Delhi, Sixth Edition, 2015
3. Introduction to Physical Metallurgy, Sidney H. Avner, McGraw Hill Education, New York; 2nd edition, 2017
4. Engineering Materials, A. K. Bhargava, Prentice Hall India Learning Private Limited, New Delhi, 2011
5. Engineering Materials – Properties and applications of Metal and Alloys, C. P. Sharma, Prentice- Hall of India Pvt. Ltd; 1st edition, New Delhi, 2004.
6. Physical Metallurgy Principles, Reed- Hill, Robert E., Reza Abbaschian, and Lara Abbaschian. 4th Ed. Cengage Learning, 2009.
7. Ashby M F, Materials Selection in Mechanical Design, Elsevier

Title	**Mechanical Measurements and Metrology**	Number	MEL2XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective: The objective of this course is to:

1. Introduce fundamental principles and standards of mechanical measurements and metrology.
2. Develop understanding of measurement instruments and error analysis.
3. Provide knowledge of limits, fits, tolerances, and geometrical measurements.
4. Expose students to advanced measurement techniques and quality control practices.

Learning Outcomes: The students are expected to have the ability to:

1. Select and use appropriate measuring instruments for engineering applications.
2. Analyze measurement errors and ensure accuracy through calibration methods.
3. Apply limits, fits, tolerances, and perform geometrical and surface measurements.
4. Utilize advanced metrology tools and interpret measurement data for quality assurance.

Contents:

1. Introduction to metrology [5 lectures]: Definition, types, need of inspection, terminologies, methods of measurement, selection of instruments, measurement errors, units, Measurement standards, calibration, statistical concepts in metrology
2. Linear metrology [4 lectures]: Steel rule, calipers, vernier caliper, vernier height gauge, vernier depth gauge, micrometers, universal caliper
3. Limits fits and tolerances [4 lectures]: Interchangeability, selective assembly, limits, fit and tolerances, limit gauging, design of limit gauges, computer aided tolerancing
4. Measurement of straightness, flatness, squareness, parallelism, roundness and cylindricity, non-contact profiling systems [5 lectures]
5. Measurement of surface finish [5 lectures]: Introduction, terminology, specifying roughness on drawings, surface roughness parameters, factors affecting surface roughness, ideal surface roughness, roughness measurement methods, precautions in measurement, surface microscopy, surface finish softwares.
6. Screw thread metrology [6 lectures]: Introduction, screw thread terminology, screw thread measurement, Gear measurement: Introduction, types of gears, gear terminology, errors in gears, advanced measurement of spur gear.
7. Miscellaneous measurements [3 lectures]: Taper measurement, angle measurement,radius measurement
8. Interferometry [3 lectures]: Principle of interference, interference bands, interference patterns, flatness interferometer, Gauge length interferometer
9. Comparator [4 lectures]: Features of comparators, classification of comparators, different comparators, advanced comparators, thread comparators.

Text/Reference books

1. K.J. Hume, Engineering Metrology, Macdonald and Co.(publisher) London
2. The Springer handbook of metrology and Testing, Czichos (Ed), 2011
3. Industrial Metrology – Smith GT, 2002, Springer

Online course material

1. https://nptel.ac.in/courses/112106179

Title	**Heat and Mass Transfer**	Number	MEL3XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	Basics of Thermodynamics & Fluid Mechanics		

Objective: The objectives of the course are:

1. To develop the ability to formulate and solve engineering problems involving conduction, convection, radiation and mass transfer.
2. To explain the design and performance analysis of fins, heat exchangers and thermal systems involving phase change.
3. To introduce empirical correlations and heat-mass transfer analogies through practical engineering applications.

Learning Outcomes: The students are expected to have the ability to:

1. Analyze steady and transient conduction problems in plane walls, cylinders, spheres and extended surfaces.
2. Evaluate heat transfer coefficients for internal/external forced convection and natural convection using appropriate correlations.
3. Analyze boiling, condensation, radiation exchange and heat exchanger performance using LMTD and effectiveness-NTU methods.
4. Apply diffusion laws and heat-mass transfer analogies to solve basic mass transfer and simultaneous heat and mass transfer problems.

Contents:

Introduction and basic concepts [1 Lectures]: Modes of heat transfer, Fourier's law, thermal conductivity, thermal diffusivity, convection coefficient, radiation heat exchange.

Heat conduction equation and boundary conditions [3 Lectures]: General heat conduction equation; one-dimensional conduction in plane walls, cylinders and spheres; boundary and initial conditions; heat generation.

Steady heat conduction [4 Lectures]: Thermal resistance concept, network analysis, composite walls/cylinders/spheres, thermal contact resistance, critical radius of insulation and heat transfer from fins.

Transient heat conduction [3 Lectures]: Lumped capacitance method, Biot and Fourier numbers, transient conduction in large plane walls, long cylinders and spheres; semi-infinite solids.

Fundamentals of convection [3 Lectures]: Velocity and thermal boundary layers, laminar and turbulent flows, conservation equations, non-dimensional numbers.

External forced convection [3 Lectures]: Drag and heat transfer in external flow, parallel flow over flat plates, flow across cylinders and spheres.

Internal forced convection [4 Lectures]: Mean velocity and temperature, entrance region, hydrodynamic and thermal entry lengths, laminar and turbulent tube flow, constant wall temperature/heat flux conditions and pressure drop.

Natural/free convection [3 Lectures]: Physical mechanism, Grashof and Rayleigh numbers, natural convection over plates, cylinders and spheres, enclosures, and mixed convection criteria.

Boiling and condensation [3 Lectures]: Boiling curve, pool boiling regimes, nucleate and film boiling, critical heat flux, dropwise and film condensation, and Nusselt's film condensation theory.

Heat exchangers [4 Lectures]: Types of heat exchangers, overall heat transfer coefficient, fouling factor, LMTD method, correction factor, effectiveness-NTU method.

Thermal radiation and radiation exchange [5 Lectures]: Radiation laws, blackbody and gray-body radiation, radiative properties, intensity, view factors, radiation exchange between black and diffuse-gray surfaces, radiation shields and electrical network analogy.

Mass transfer [3 Lectures]: Heat-mass transfer analogy, Fick's law, diffusion in stationary and moving media, boundary conditions, steady and transient mass diffusion.

Text/Reference books

1. Çengel, Y. A. and Ghajar, A. J., Heat and Mass Transfer: Fundamentals and Applications, McGraw-Hill Education.
2. Incropera, F. P., DeWitt, D. P., Bergman, T. L. and Lavine, A. S., Fundamentals of Heat and Mass Transfer, John Wiley & Sons.
3. Sukhatme, S. P., A Textbook on Heat Transfer, Universities Press.
4. Holman, J. P., Heat Transfer, McGraw-Hill Education.

Online course material

https://nptel.ac.in/courses/112101097

https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_ch10

Title	**Kinematics and Dynamics of Machines**	Number	MEL2XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	–		

Objectives

1. To cover the kinematics and dynamics of planar single degree freedom mechanisms
2. To develop skills for designing and analyzing linkages, cams, gears and other mechanisms
3. To teach students concepts of Newtonian dynamic analysis of mechanisms and machines

Learning Outcomes

1. To analyze position, velocity and acceleration kinematics of mechanisms
2. Evaluation of forces and torques in mechanism and machines in operation
3. To understand the implication of computed results to improve design of machines

Contents:

Kinematics and Mechanism [2 Lectures]: Links, Kinematic pairs, Kinematics diagram and inversion, mobility, range of movements, Equivalent linkage

Motion characteristics of Mechanism [5 Lectures]: Displacement analysis of planar mechanisms analytical methods, Plane motion of a rigid body, Velocity analysis: analytical and Instantaneous Centre of Velocity methods, Coriolis component of acceleration, Acceleration analysis

Mechanism Synthesis [4 Lectures]: Function, Path, and Motion Generation, Three Position Dimensional Synthesis

Cam follower mechanism [4 Lectures]: Classification of Cam and followers, nomenclature, description and analysis of follower motion, pressure angle, Determine of basic dimensions and synthesis of cam profiles

Gear and gear train [5 Lectures]: Spur gears and gear terminology, Fundamental law of gearing, Involute profile, Interference and undercutting, Minimum number of teeth, Contact ratio, Helical gears on parallel shafts, Bevel helical, Spiral and Worm gears. Simple, compound, epicyclic gear trains, planetary gear trains.

Introduction to Dynamics [2 Lectures]: Dynamics lumped-parameter models of mechanical systems, Newton's Law, Inertia and D Alembert Principle, Energy Method

Dynamics of Rigid bodies [5 Lectures]: Solution Methods: superposition and simultaneous equation solution, Newtonian Approach, Shaking force and Shaking Moments, Applications to Linkage and Single and Multi-cylinder engines, Controlling Input Torque - Flywheel Design

Balancing [4 Lectures]: Static unbalance, static balancing machines, dynamics balancing, applications to balancing of linkage, single and multi-cylinder engines

Vibration Analysis [4]: Introduction, Basic features of vibratory systems, single degree freedom systems

Dynamics of Cams [3 Lectures]: Rigid and elastic body cam system, analysis of eccentric cams, analysis of elastic cam systems, lumped models of cams and vibration analysis.

Gyroscope [1 Lecture]: Gyroscopic Torque

Text book

1. Norton, R. L., (2009), Kinematics and Dynamics of Machinery, Tata McGraw Hill

Reference books

1. Ghosh, A. & Mallik, A. K., (2009), Theory of Mechanisms, and Machines, 3rd Edition, East West Press
2. Uicker, J. J., Pennock, G. R. and Shigley, J. E., (2009), Theory of Machines and Mechanisms, 3rd Edition, Oxford University Press
3. Waldron, K. J. & Kinzel, G. L., (2004), Kinematics, Dynamics and Design of Machinery, 2nd Edition, Wiley
4. Thomson, W. T., Theory of Vibrations with Applications, CBS Publishers & Distributors
5. Paul, B., Kinematics and Dynamics of Planar Mechanisms', Prentice Hall, 1979.

Self-learning

1. Ashok K Mallik, Kinematics of Machines, NPTEL Course, Department of Mechanical Engineering, IIT Kanpur, Link: https://nptel.ac.in/courses/112104121/
2. Amitabh Ghosh, Dynamics of Machines, NPTEL Course, Department of Mechanical Engineering, IIT Kanpur, Link: https://nptel.ac.in/courses/112/104/112104114/

Title	**Machining Science**	Number	MEL3XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objectives:

1. Impart fundamental understanding of various conventional and non-conventional machining processes.
2. Discuss various measuring and gauging techniques used for inspection of machined components.
3. Discuss various techniques for improvement and assessment of the surface finish.

Learning Outcomes:

1. Appreciate theories of material removal associated with conventional and non-conventional machining processes.
2. Understand various aspects related with finishing and super-finishing processes.
3. Apply different inspection techniques to assess the dimensional accuracy of machined components.

Contents:

Conventional Machining Processes [17 Lecture]: Introduction [3 Lecture]: Machining Process; Basic working principle of conventional machining processes, Working principle and construction of lathe, Types of lathe, Tool and Chip formation [5 Lecture]: Configuration, Specification and Classification of machine tools; Mechanics of Metal Cutting: Geometry of single point cutting tools; Mechanism of chip formation; Orthogonal and oblique cutting. Force and temperature analysis [6 Lecture]: Analysis of machining forces and power requirement; Causes, effects and control of cutting temperature; Application of cutting fluids. Machinability and tool life [3 Lecture]: Machinability concept; Tool Wear; Failure of cutting tools and tool life; Cutting Tool Materials; Economics of metal cutting.

Finishing process [5 Lecture]: Finishing and Super-Finishing Processes: Purpose and application of grinding; Selection of grinding wheels and their conditioning methods; Specific energy and power requirement in grinding; Super-finishing processes such as Honing, Lapping etc.

Advanced Machining Processes [17 Lecture]: Introduction [2 Lecture]: Comparison of non-conventional and conventional methods of machining; Classification of the processes; Advantages, limitations and applications. Mechanical Processes [7 Lecture]: Abrasive Jet Machining Process, process parameters effect on material removal rate, Water Jet Machining (WJM) process; Abrasive Water Jet Machining (AWJM) process; Ultrasonic Machining (USM), Material removal in USM. Thermo-electric Processes [6 Lecture]: Electric Discharge Machining (EDM); Use of LASERs in machining and Laser Beam Machining, Electron Beam Machining (EBM) and Plasma Machining (PM). Electrochemical Processes [2 Lecture]: Electroplating and reverse-electroplating; Electro Chemical Machining (ECM).

Text book:

1. Shaw, M C, (2014), Metal Cutting Principles, Oxford University Press.
2. McGeough, J A, (1988), Advanced Methods of Machining, Springer.
3. Mitutoyo, (2015), Metrology Handbook, Mitutoyo.

Reference Books:

1. Chattopadhyay, A B, (2013), Machining and Machine Tools, Wiley India.
2. Galyer, J F W, Shotbolt, C R, (1990), Metrology for Engineers, Cassell.
3. Boothroyd, G., and Knight, W. A., (1932), Fundamentals of Machining and MAchine Tools, CRC Press.

Self-Learning Material:

1. V. K. Jain, Advanced Machining Processes, NPTEL Course Department of Mechanical Engineering, IIT Kanpur, Link: http://nptel.ac.in/courses/112104028/
2. M. S. Shunmugam, Metrology, NPTEL Course Department of Mechanical Engineering, IIT Madras, Link: http://nptel.ac.in/courses/112106138/.
3. U. S. Dixit, Mechanics of Machining, NPTEL Course Department of Mechanical Engineering Guwahati, Link: http://nptel.ac.in/courses/112103248/
4. A. B. Chattopadhyay, Manufacturing Processes II, NPTEL Course of Department of Mechanical Engineering, IIT Kharagpur, https://nptel.ac.in/courses/112/105/112105126/

Title	**Introduction to Smart Automation**	Number	MEL3XXX
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech	Type	Core
Prerequisite			

Objectives: The objectives of the course are;

1. To introduce the concepts of smart and connected automation systems,
2. To explain elements enabling the transition from standalone automation to integrated information-driven systems.
3. To develop an understanding of industrial communication, supervisory systems, intelligent monitoring, and digital automation in smart factories.

Learning Outcomes: The course will enable students to achieve the following;

1. Understand and analyze smart and integrated automation systems.
2. Apply concepts of industrial communication, intelligent monitoring, and digital automation in modern industrial systems.

Course Contents

Smart Automation and Integrated Industrial Systems: Evolution of industrial automation, Standalone vs integrated automation systems, Automation hierarchy, Cyber-physical systems, Industry 4.0 and Industry 5.0 concepts, Smart factories, Flexible automation cells, System-level integration, Machine connectivity and interoperability, Digital twin concepts. [6 lectures]

Industrial Communication and Supervisory Automation Systems: Industrial communication architecture, Industrial Ethernet, Modbus TCP/IP, PROFINET, EtherCAT, OPC-UA, MQTT, Machine-to-machine communication, Edge-to-cloud connectivity, SCADA systems, HMI systems, Supervisory control, Production monitoring, OEE monitoring, Web-enabled dashboards, Real-time system visibility, Supervisory digital twins, and real-time industrial data synchronization. [8 lectures]

Integrated Material Handling and Inspection Automation: Automated material handling systems- Conveyors, AGVs, AMRs, ASRS, Warehouse automation- Buffering systems, CNC-robot integration, In-line inspection systems- On-machine inspection, CMM integration, machine vision systems, process-quality integration, closed-loop inspection systems, CAD/CAM integration, digital manufacturing chain, PLM concepts, digital Thread, Digital Shadow, and Digital Twin. [10 lectures]

Intelligent Monitoring, Predictive Maintenance, and Virtual Automation Systems: Condition monitoring, Fault diagnosis, Predictive maintenance, Vibration and energy monitoring, Tool and equipment health monitoring, Data-driven diagnostics, AI-assisted industrial monitoring, Digital twin-driven monitoring systems, Digital factory concepts [7 lectures]

Blockchain and Smart Enterprise Automation Systems: Introduction to blockchain, Distributed ledger concepts, Blockchain architecture, Smart contracts, Blockchain-enabled industrial communication, Blockchain for machine connectivity and IIoT systems, Blockchain-enabled production and supply chains, Cybersecurity aspects, Challenges and limitations of blockchain in industrial automation [8 lectures]

Reference Books

1. Groover M. P., *Automation, Production Systems, and Computer-Integrated Manufacturing*, Pearson.
2. Kusiak A., *Smart Manufacturing Systems*, Wiley
3. Mahalik N. P., *Smart Sensors and Internet of Things*, Springer.

Self-Learning Material

1. Introduction to Industry 4.0 and Industrial IoT by Prof. Sudip Misra - https://onlinecourses.nptel.ac.in/noc24_cs34/preview

Title	**Power Systems Engineering**	Number	MEL3XXX
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective:

1. To understand fundamentals of thermodynamics, fluid mechanics, combustion, and energy conversion systems.
2. To study the operation and performance of IC engines, turbomachines, and power plants.

Learning Outcomes:

1. Students will be able to analyze engines, compressors, turbines, and power cycles using thermodynamic principles.
2. Students will be able to evaluate performance, efficiency, and emissions of modern energy systems.

Contents:

Introduction[2 Lectures]: Basics of Fluid Mechanics and Thermodynamics Laws

Fuels and Combustion [3 Lectures]: Introduction to combustion, Reacting systems, Stoichiometry and exhaust gas composition, Heat of formation, Adiabatic Flame Temperature

IC Engines [7 Lectures]: Introduction to SI and CI engines, Air standard cycle analysis, 2-S and 4-S engines, SI & CI engine combustion, Performance and testing of SI and CI engines, IC engine emissions, after treatment systems, Low carbon propulsion Techniques

Gas Turbine Power Plants [5 Lectures]: Layout of gas turbine power plant, Brayton Cycle, with reheat, intercooling, regeneration.

Thermal Power Plants [5 Lectures]: Steam power plant cycle, reheating, regeneration, co-generation, Layout of a thermal power plant, classification of steam generators

Turbomachinery Basics [8 Lectures]: Classifications, Eulers equation of Turbomachines, Concept of velocity triangles, Expression of Eulers equation showing the components of energy transfer. Introduction to compressible flows, flow choking, review of T-S and H-S diagrams of gas turbines and compressors, efficiencies, effect of reheat and preheat. Dimensional analysis of compressible flow machines. Lift and drag, blade cascading concept

Compressors and Turbines [9 Lectures]: Centrifugal compressors, construction, flow analysis, performance analysis, stalling and surging, Axial flow compressors, construction details, stage velocity triangles, degree of reaction, stage pressure rise, stage pressure coefficient, stage efficiency, Axial Turbines,Reaction stages-50% and 100% reaction

Text/Reference books

1. Moran, M. J., Shapiro, H. N., Boettner, D. D., & Bailey, M. B. (2018). Fundamentals of engineering thermodynamics (9th ed.). Wiley.
2. Pulkrabek, W. W. (2004). Engineering fundamentals of the internal combustion engine (2nd ed.). Pearson Prentice-Hall.
3. Dixon, S. L., & Hall, C. A. (2014). Fluid mechanics and thermodynamics of turbomachinery (7th ed.). Butterworth-Heinemann.
4. Korpela, S. A. (2020). Principles of turbomachinery (2nd ed.). John Wiley & Sons.
5. Nag, P. K. (2021). Nag's power plant engineering (5th ed.). McGraw Hill Education.

Title	**Computer-assisted Fabrication**	Number	MEL3XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective: The objective of the course is

- To develop proficiency in digital design and modeling, including CAD and geometric modeling.
- To develop understanding of CNC programming and additive manufacturing technologies.
- To introduce digital workflows and Industry 4.0 concepts in manufacturing.

Learning Outcomes: The students are expected to have the ability to:

- Create and prepare digital models for fabrication using CAD tools and appropriate file formats.
- Analyze and select suitable manufacturing processes for engineering applications.
- Explain CNC programming, additive manufacturing, and Industry 4.0 concepts.

Contents:

Digital Design & Modeling (14 Lectures)

Introduction to digital fabrication, Computer-Aided Design (CAD) fundamentals (solid, surface, parametric modeling), Geometric modeling (curves, surfaces, solids), File formats (STL, AMF), tessellation, 3-D object scanning, Digital workflow & process chain

Digital manufacturing (20 Lectures)

Subtractive Manufacturing (SM): Fundamentals of machining, CNC Machine structure & drives, CNC control systems & axes, fundamentals of CNC programming, CNC programming for machining centers and turning centers, Canned cycles, metrology & inspection in CNC

Additive Manufacturing (AM): Introduction to AM technologies and classification, materials for AM, defects and quality Control , capabilities and limitations, various applications of AM.

Introduction of Industry 4.0 (5 Lectures)

Fundamentals of Industry 4.0, Internet of Things (IoT), Cyber-Physical Systems (CPS), applications, case studies, and future Trends

Text/Reference books

1. Zeid, I and Subramanian R, (2010), CAD/CAM Theory and Practice, Tata McGraw Hill
2. Overby, A., CNC Machining Handbook: Building, Programming, and Implementation, 1st ed. New York, NY, USA: McGraw-Hill, 2010.
3. Smid, P., CNC Handbook, 3rd ed. New York, NY, USA: McGraw-Hill, 2008.
4. Gibson I, Rosen D W, Stucker B, (2015), Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing, Springer

Online course material

https://nptel.ac.in/courses/112102101
https://nptel.ac.in/courses/112105211
https://onlinecourses.nptel.ac.in/noc26_mm14/preview

Title	**Production & Operations Management**	Number	MEL3XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [0]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective: The objective of the course is:

1. Familiarize students with basic concepts, techniques and algorithms that help in optimizing the performance of manufacturing systems

Learning Outcomes: The students are expected to have the ability to:

1. Appreciate the need of developing mathematical models of conflicting situations and analyze the same.
2. Select rational options in practical decision-making problems using various techniques and algorithms.

Contents:

Introduction (15 Lectures): Introduction [2 Lecture]: System concept of manufacturing; Types of manufacturing systems; Concept of productivity. Linear Programming [7 Lecture]: Introduction to Linear Programming Problem (LPP); Formulation of mathematical model; Applications of LPP; Solution techniques for LPP such as graphical method, simplex method and its variants; Special issues in LPP e.g. degeneracy, multiple optimum solutions, infeasibility, unbounded feasible region; Sensitivity analysis. Transportation Problem and Assignment Problem [6 Lecture]: Transportation problem and its applications; Solution methods for transportation problem; Transhipment problem; Assignment problem and its applications; Solution techniques for assignment problem.

Inventory Management (14 Lectures): Introduction [3 Lecture]: Basic concept and terminology; Inventory classification and inventory costs; Concept of Economic Order Quantity. Deterministic inventory models [3 Lecture]: Probabilistic inventory models; ABC analysis. Project Management [1 Lecture]: Introduction and terms used in project management; Construction of network diagram. PERT/CPM [4 Lecture]: Deterministic network model - CPM; Probabilistic network model-PERT; Crashing of network; Resource leveling and smoothing techniques. Waiting Line Models [3 Lecture]: Introduction and terms used in queuing models; Kendall's notations; Classification of queuing models; Mathematical analysis of queuing system using various models; Queuing costs.

Forecasting (10 Lectures): Forecasting [4 Lecture]: Nature and use of forecast; various forecasting models; Measure of forecasting accuracy; Selection of forecasting techniques. Work study [4 Lecture]: Purpose and scope, Productivity and work-study, Method Study and Work Measurement; Engineering economy and costing: Fixed cost, variable cost, introduction to break-even analysis, Sequencing [2 Lecture]: Objectives, Johnson's Rule, Types of Sequencing Problems.

Text books:

1. Buffa E S and Sarin R K, (1987), Modern Production/Operations Management, Wiley.
2. Taha H A, (2010), Operations Research: An Introduction, Pearson.

Reference Books:

1. Sen R, (2016), Operations Research, Prentice Hall.
2. Hiller F S and Liberman G J, (2005), Introduction to Operations Research, McGraw Hill.
3. Pai J K, (2010), Operations Research, Oxford University Press.

Self-Learning Material:

1. G Srinivasan, Operations Research, NPTEL Course Department of Mechanical Engineering, IIT Madras, Link: http://nptel.ac.in/courses/112106134/1
2. A Kanda, Project and Production Management, NPTEL Course Department of Mechanical Engineering, IIT Delhi, Link: http://nptel.ac.in/courses/112102106/

Title	**Control of Mechanical Systems**	Number	MEL3XX0
Department	Mechanical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B. Tech (ME)	Type	Program Compulsory (PC)
Prerequisite	Basic Knowledge of Differential Equations and Laplace Transform		

Objectives

1. To familiarize students with methods to characterize the dynamical behaviour of mechanical systems using mathematical models.
2. To provide an understanding of the fundamental concepts of control theory.

Learning Outcomes

1. Students will be able to develop mathematical models of mechanical systems and apply appropriate methods for their representation and analysis.
2. Students will be able to design control systems for linear time-invariant (LTI) systems.

Contents

***Introduction to Control [3 Lectures]*:** Control objectives; introduction to control problems; open- and closed-loop control structures; properties of feedback systems; basic sensitivity analysis; effect of control gain on system performance; effect of disturbances on tracking error.

***Introduction to Modelling & Representation [6 Lectures*]:** Objectives; basic modelling concepts and types of models; models of simple mechanical systems; linearization of nonlinear models; LTI systems; transfer function and state-space modelling of LTI systems; conversion from state space to transfer function; block diagram representation and manipulation.

***Response & Stability Analyses [5 Lectures]*:** Test signals; response of first- and second-order as well as higher-order systems; poles and zeros and their effect on system response; stability criteria; BIBO stability; Routh's stability criterion.

***Analysis of Feedback System [4 Lectures]*:** Basic control equation: stability, regulation, tracking, and sensitivity; control of steady-state errors to polynomial inputs; system type and steady-state error; proportional (P), integral (I), and derivative (D) control; PI and PID control; Ziegler–Nichols tuning of the PID controller.

Root Locus and Compensator Design [6 Lectures]: Concept of root locus; use of control gain to change pole locations; design of lead, lag, and lead–lag compensators using root locus techniques.

Frequency Response Design Method [6 Lectures]: Frequency response and its representation using Bode and Nyquist plots; Nyquist stability analysis; concepts of gain and phase crossover frequencies; gain and phase margins; design of PD, PI, lead, lag, and PID compensators.

Design in State Space [5 Lectures]: Canonical forms of state equations; response from state equations; control law design for full-state feedback; selection of poles; estimator design; combined control law and estimator.

***Implementation of Control Law [2 Lectures*]:** Discretization of control law; difference equations; computational aspects of control law.

***Applications of Control to Mechanical Systems [2 Lectures]*:** Practical aspects of control implementation—actuators and sensors; temperature control; flow control; pressure and force control; vehicle active suspension control; robot control.

Textbook:

1. Franklin, G. F., Powell, J. D. and Emami-Naeini, A. E. (2022), Feedback Control of Dynamic Systems, 8th Edition, Pearson India

Reference Books

1. Gopal, M. (2000), Control Systems - Principles & Design, 3rdEdition, Tata McGraw-Hill, 2015
2. Dorf, R.C. & Bishop, R.H. (2008), Modern Control Systems, 11th Edition, Pearson, 2009
3. Ogata, K. (2012), Modern Control Engineering, 5th Edition, Prentice-Hall India, 2012,
4. Nise, N.S. (2014), Control Systems Engineering, 6th Edition, John-Wiley India, 2014

Self-Learning Material

1. Gopal, M., Control Engineering, NPTEL Course Material, Department of Electrical Engineering, Indian Institute of Technology Delhi, https://nptel.ac.in/courses/108102043/
2. Ramkrishna Pasumarthy R., Control Engineering, NPTEL Course Material, Department of Electrical Engineering, IIT Madras, Link: https://onlinecourses.nptel.ac.in/noc17_ee12/preview

Title	**Applied Thermofluids**	Number	MEL3XXX
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Objective:

1. To understand the fundamentals of refrigeration and air conditioning principles and methods, and performance analysis of different refrigeration/cooling cycles
2. To understand psychrometry and the psychrometric process for different heating, ventilation, and air conditioning systems
3. To design a refrigeration/cooling system for industrial application, case studies/field visits, projects, including natural refrigerants
4. To study the fundamentals and performance of turbomachines, hydraulic turbines, and centrifugal pumps.

Learning Outcomes:

1. Understand and evaluate the performance of different refrigeration systems
2. Design a basic refrigeration cycle/system for different applications, such as thermal comfort, cold chain, and transport refrigeration, etc.
3. Evaluate the performance of different psychrometric processes
4. Compute the refrigeration plant's cooling load/capacity and select components.
5. Students will be able to evaluate the operation and characteristics of turbines, pumps, and related fluid machinery.

Contents:

Basic concepts [2 Lectures]*:* Need for refrigeration, basic thermodynamics, and heat transfer, Methods of refrigeration, refrigeration and heat pump cycles.

Vapour compression refrigeration cycle [4 Lectures]: Carnot refrigeration cycle, Simple VCRC, elements of VCRS, compressors, evaporator, condenser, expansion valves, actual VCRS, design calculation, performance improvement, multistage and cascade refrigeration systems.

Refrigerants and environmental impact [3 Lectures]: Refrigerant classification and nomenclature, thermophysical properties, performance mapping, environmental impact, and policies.

Psychrometry [4 Lectures]: Properties of moist air, psychrometry, sensible and latent cooling load, apparatus dew point, psychometric chart, air conditioning processes, Mixing process, Basic processes in conditioning of air, evaporative coolers and cooling towers.

Load Calculations [4 Lectures]: Outside and inside design conditions, Sources of heating and cooling load, Heat transfer through structure, Solar radiation, Electrical applications, Infiltration and ventilation, Heat generation inside conditioned space, Human physiology, Comfort chart.

Non-conventional refrigeration systems [3 Lectures]: Sorption refrigeration, Ejector refrigeration, Vortex refrigeration, Dew-point evaporative cooling-Introduction, Analysis, Relative merits and demerits, Performance applications.

Industrial application/field visits [3 Lectures]: Central Cooling plant, Cold storage, Ice plants, Supermarket, VRFs -Type of compressors and their performance curves, Types of condensers, Heat transfer in condensers, Types of expansion devices, Types of evaporators, Cooling and dehumidifying coils.

Special Topics/Projects [3 Lectures]: Guest lectures, Natural refrigerant-based cooling system, off-grid solar-powered cold room - Classifications, Layout of plants, Equipment selection, Air distribution system, Temperature sensors, Pressure sensors, Humidity sensors, Actuators, Safety controls, Accessories.

Incompressible Turbomachines [5 Lectures]: Classifications, Eulers equation of Turbomachines, Concept of velocity triangles, Expression of Eulers equation showing the components of energy transfer, Similarity laws and model testing., concept of specific speed and specific diameter, affinity laws, unit quantities.

Hydraulics Turbines [4 Lectures]: Impulse and reaction machines, Analysis of Pelton-Turbine, efficiencies, performance characteristics, Analysis of Francis Turbine, Analysis of Kaplan Turbine.

Centrifugal Pumps [4 Lectures]: Introduction to pumping systems, Centrifugal pumps working principle, Efficiencies, losses, operating characteristics, pumps in series and parallel, NPSH, priming and cavitation.

Text/Reference books

1. Jordan, R. C. & Priester, G. B., (2003), Refrigeration & Air conditioning, Prentice Hall of India
2. Arora, C. P., (2008), Refrigeration & Air conditioning, Tata McGraw Hill
3. Stocker, W. F. & Jones, J. W., (2000), Refrigeration & Air conditioning, Tata McGraw Hill
4. Dixon, S. L., & Hall, C. A. (2014). Fluid mechanics and thermodynamics of turbomachinery (7th ed.). Butterworth-Heinemann.
5. Korpela, S. A. (2020). Principles of turbomachinery (2nd ed.). John Wiley & Sons.
6. ASHRAE Handbooks/Manuals.

Online course material

https://nptel.ac.in/courses/112105128

Title	**Mechanics and Manufacturing Lab**	Number	MELP2XX0
Department	Mechanical Engineering	L-T-P [C]	0-0-6 [3]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Part-I: Kinematics and Dynamics of Machines Lab

Objectives

1. To cover the kinematics and dynamics of planar single degree freedom mechanisms.
2. To develop skills for designing and analyzing linkages, cams, gears and other mechanisms.
3. To teach students concepts of Newtonian dynamic analysis of mechanisms and machines.

Learning Outcomes

1. To analyze position, velocity and acceleration kinematics of mechanisms.
2. Evaluation of forces and torques in mechanism and machines in operation.
3. To understand the implication of computed results to improve design of machines.

Contents:

Simulation-based experiments (MATLAB/ ADAMS)

1. Create linkages and gear trains in ADAMS and perform their motion analysis.
2. Analytical position, velocity and acceleration analysis of four-bar mechanisms in MATLAB.
3. Analytical position, velocity and acceleration analysis of the slider-crank mechanism in MATLAB.
4. Graphical position and velocity analysis of mechanisms with fixed sliders/ four-bar mechanisms.
5. Graphical acceleration analysis of mechanisms with fixed-sliders/ four-bar mechanisms.
6. CAM Design: plot the cam profile for the given rise, dwell, and fall sections.
7. Perform three-position dimensional synthesis of a four-bar mechanism.
8. Perform force analysis of a mechanism using the Newtonian solution approach.

Hardware-Based Experiments

9. Inversions and Grashof's criteria: to construct inversions of a four bar mechanism and to study the mechanism motion qualitatively.
10. Vibration (Cantilever Beam): Find out the undamped natural frequencies of a cantilever beam.
11. Motorised Gyroscope: to study the effect of an external torque on the precession speed of the gyroscope.
12. CAM Analysis: Plot displacement vs angle, study the effect of follower weight and cam speed on bounce.
13. Static and Dynamic Balancing Apparatus: to balance the masses statically and dynamically of a simple rotating mass system and to observe the effect of unbalance in a rotating mass system.

Text book

1. Norton, R. L., (2009), Kinematics and Dynamics of Machinery, Tata McGraw Hill

Reference books

1. Ghosh, A. & Mallik, A. K., (2009), Theory of Mechanisms, and Machines, 3rd Edition, East West Press
2. Uicker, J. J., Pennock, G. R. and Shigley, J. E., (2009), Theory of Machines and Mechanisms, 3rd Edition, Oxford University Press

Part-II: Fluid Mechanics Lab

Objectives:

1. To introduce various forms of flow measuring devices and techniques
2. To provide hands on experience on in the area of experimental fluid mechanics

Learning Outcomes: The students are expected to have the ability to:

1. Design experimental methods for fluid flow problems
2. Interpret data from experiments of fluid flows and usage of charts and hand books

Contents (12-13 Sessions):

1. Determination of head loss in pipes having different diameters, different materials and different roughness
2. Determination of pressure drop at different types of bends and valves
3. Reynolds apparatus to measure critical Reynolds number for pipe flows

4. Measurement of Cp distribution on a cylinder
5. Wake survey experiments in a flow over a cylinder
6. Impact of jet on flat and curved plates
7. Wind tunnel calibration using Pitot static tube
8. Working principle of different flow meters (orifice plate, venture meter, turbine, Rota meter, electromagnetic flow meter)
9. Effect of change in cross section and application of the Bernoulli equation
10. Flow measurement around an object or blockage using PIV techniques
11. Class projects for designing and measurement of simple fluid mechanics experiments

Reference books

1. Fox, R. W., Pitchard, P. J and McDonald, A. T., (2010), Introduction to Fluid Mechanics, 7th Edition, John Wiley & Sons Inc.
2. Kundu, P. K, Cohen, I. M and Dowling, D. R, (2014), Fluid Mechanics, 6th Edn., Associated Press
3. White, F. M., (2003), Fluid Mechanics, 5thEdition, McGraw-Hill

Online course material

1. S. Chakrabarty, Introduction to Fluid Mechnics, NPTEL Course Material, Mechanical Engineering, IIT Kharagpur, https://nptel.ac.in/courses/112/105/112105269/
2. V.Shankar, Fluid Mechanics, NPTEL Course Material, Chemical Engineering, IIT Kanpur, https://nptel.ac.in/courses/103/104/103104044/#
3. Nishith Verma, Fluid Mechanics, NPTEL Course Material, Chemical Engineering, IIT Kanpur, https://nptel.ac.in/courses/103/104/103104043/

Part-III: Manufacturing Processes Lab

Objective: The objective of the course is:

1. Demonstrate concepts related with primary manufacturing processes by conducting various practical exercises.
2. Familiarize students with various simulation tools involved in manufacturing processes and systems.
3. Explain the software tools and skills required for optimization of manufacturing systems.

Learning Outcomes: The students are expected to have the ability to:

1. Appreciate the effect of process parameters on quality of manufactured components.
2. Use various simulation tools and software for designing manufacturing processes and systems.

Contents:

1. Studying the effect of the clay and moisture content on sand mold properties.
2. Derivation of cooling curve and mold thermal gradient using thermocouple and data acquisition system.
3. Studying the effect of welding process parameters on bead profile using image processing system.
4. Non-destructive testing of the cast and weld components.
5. Analysing the problem of heat balance using resistance spot welding setup.
6. Effect of interface lubrication and temperature on bulk deformation characteristics using hydraulic press.
7. Sheet metal punch/die design and layout optimization.
8. Solving linear programming problems using LINGO software.
9. Demonstration of material flow and solidification simulation using Auto-Cast software.
10. Derivation of isotherms and estimation of bead profile using Ansys Software.
11. Development of computational tool using Microsoft excel for solving transportation/Assignment and Inventory management problem.
12. Effect of process parameters in GTAW process through microstructural analysis of weld bead Evaluation.
13. Study of the mechanical properties of the cast and weld components.

Text/Reference books:

1. Laboratory manual of various equipment.
2. Reveillac J, (2017), Operational research using the spread sheet. Wiley.

Title	**Mechanical Measurements and Characterization Lab**	Number	MELP2XX0
Department	Mechanical Engineering	L-T-P [C]	0-0-8 [4]
Offered for	B.Tech.	Type	PC
Prerequisite	--		

Part-I: Heat and Mass Transfer Lab

Objective: The objectives of the course are:
1. To provide hands-on exposure to conduction, convection, radiation, phase change and mass transfer.
2. To train students in thermal measurements, sensor calibration, heat exchanger testing and mass transfer coefficient estimation.
3. To connect experimental data with governing laws, correlations and design calculations.

Learning Outcomes: The students are expected to have the ability to:
1. Conduct experiments related to heat and mass transfer processes.
2. Calibrate temperature sensors and estimate measurement uncertainty.
3. Interpret data using Fourier's law, Newton's law of cooling, radiation laws, LMTD/effectiveness methods and Fick's law.
4. Prepare concise reports comparing measured results with theory and standard correlations.

Contents:

Exp. No.	Name of experiment
1	Temperature distribution during 1-D heat transfer in plain and composite walls.
2	Demonstration of domestic solar water heating system: time vs temperature plots.
3	Measurement of thermal conductivity of an unknown material.
4	Study of effect of solar luminous intensity, water flow rate, and angle of incidence on collector performance.
5	Study of forced and free convection heat transfer over flat plate, tube bundle, and fins; comparison with correlations.
6	Temperature sensors: K-type thermocouple and PT100 calibration, accuracy and time-response study.
7	Radiation laws: Lambert's distance law, Lambert's cosine law, and Stefan–Boltzmann law.
8	Heat exchangers in parallel-flow and counter-flow arrangements using LMTD/effectiveness concepts.
9	Pool boiling experiment: boiling curve, onset of nucleate boiling and critical heat flux trend.
10	Unsteady-state heat transfer and estimation of transient thermal parameters.
11	Determination of diffusion coefficient of vapour in air using vapour diffusion apparatus.
12	Study of diffusion from solid/vapor source into air and estimation of diffusion coefficient or mass transfer rate.

Text/Reference books

1. Cengel, Y. A. and Ghajar, A. J., Heat and Mass Transfer: Fundamentals and Applications, McGraw-Hill Education.
2. Incropera, F. P., DeWitt, D. P., Bergman, T. L. and Lavine, A. S., Fundamentals of Heat and Mass Transfer, John Wiley & Sons.
3. Holman, J. P., Heat Transfer, McGraw-Hill Education.

Online course material
https://nptel.ac.in/courses/112101097
https://ht-iitb.vlabs.ac.in/
https://archive.nptel.ac.in/courses/112/105/112105271/

Part-II: Experimental Solid Mechanics Lab

Objective: To provide hands-on understanding of stress-strain behavior, deformation, and failure of materials through experimental methods.

Learning Outcomes: The students are expected to have the ability to:
1. Experimentally determine the elastic and strength properties of materials.
2. Analyze structural response under axial, torsional, and bending loads.
3. Validate classical theories such as Euler–Bernoulli beam theory and column buckling.
4. Use strain measurement techniques for stress analysis.

Contents:

1. Uniaxial tensile test to determine elastic constants
2. Engineering stress-strain and true stress-strain.
3. Evaluation of Poisson's ratio on printed specimens.
4. Compression test of a 3D lattice.
5. Torsion test to determine elastic constants.
6. Characterization of fracture surface and verification of failure plane for brittle and ductile systems.
7. Beam bending test to verify the Euler-Bernoulli beam theory.
8. Stress measurement in a thin pressure vessel.
9. Beam bending to find deflection and stress.
10. Application of photoelasticity to measure stress concentration.
11. Buckling test.
12. Demo of DIC.

Text/Reference books

1. Dieter, G.E. and Bacon, D., 1976. Mechanical metallurgy (Vol. 3, pp. 43-53). New York: McGraw-hill.
2. Crandall, S.H., Dahl, N.C., and Lardner, T. J., (1978), An Introduction to the Mechanics of Solids, McGraw Hill, Second Ed. with SI Units

Online course material
https://nptel.ac.in/courses/112103167
https://open.umich.edu/find/open-educational-resources/engineering/lectures-continuum-physics

Part-III: Engineering Materials and Processing Techniques Lab

Objective:
The objective of the lab is
- To familiarize students with hands-on experience associated with basic manufacturing processes and the preparation of test specimens for materials characterization.
- To develop practical understanding of microstructural analysis and mechanical property evaluation using standard characterization techniques.

Learning Outcomes:
The students are expected to have the ability to:
- Manufacture simple components and prepare material samples for metallographic and mechanical characterization.
- Analyze microstructure, measure grain size and hardness, and relate processing conditions to material properties.

Contents:
1. Fabrication of the components using primary manufacturing processes.
2. NDT testing of the fabricated components (LPT and Ultrasonic testing).
3. Sample cross-sectioning for the metallography and mechanical testing.
4. Macro and micro level characterization of the components using stereoscope and an optical microscope.
5. Mechanical testing of the components (Charpy impact, and Tensile testing).

6. To determine the Rockwell hardness number and Brinell hardness number for a given metallic specimen.
7. To determine the hardness number for a given metallic specimen by Vickers hardness testing
8. Residual stress measurement in the manufactured components.
9. Study on various heat treatments for mild steel.
10. Study on the role of heat treatment on the microstructure-mechanical property relationship for mild steel.
11. Microstructural study of non-ferrous alloys.
12. Scratch testing of the manufactured components.

Reference books:

1. W.D. Callister, Introduction to Materials Science, 2010
2. V. Raghvan, Introduction to Materials Science, 2015

Part-IV: Mechanical Measurements and Metrology Lab

Objective:

- To impart practical knowledge of dimensional measurement, geometric inspection, calibration, and operation of basic measurement instruments and transducers.
- To familiarize students with measurement standards, error sources, and industrial inspection practices in mechanical engineering applications.

Learning Outcomes:

- Students will be able to carry out precision measurements, calibrate instruments, and analyse measurement data considering accuracy and uncertainty.
- Students will be able to select suitable measurement techniques and instruments for dimensional inspection, process monitoring, and engineering experimentation.

Experiments:

1. Measurement of Linear Dimensions using Vernier Caliper and Micrometer & Circular Run-out using Dial Indicator.
2. Measurement of Step Height and Surface Flatness using a Height Gauge and Surface Plate, and Calibration of an Outside Micrometer using Slip Gauges (Gauge Blocks).
3. Analysis of surface topography using white light interferometry (WLI)
4. Measurement of Strain in a Cantilever Beam using Electrical Resistance Strain Gauge.
5. Calibration and Force Measurement using Load Cell or dynamometer.
6. Measurement of Surface Flatness using Optical Flat and Monochromatic Light source.
7. Measurement of Tool Nose Radius and Tool Angles using Tool Maker's Microscope.
8. Measurement of Temperature using Thermocouple and Infrared gun.
9. Quantitative Measurement of Internal Flaws in Metallic Plates using Phase-array UT.
10. Measurement of Vacuum Leaks using pressure rise/helium leak detection method.
11. Surface texture characterization using advanced surface texture parameters (WLI)
12. Surface defect identification using magnetic particle inspection (MPI) technique

Reference books

[1]. E. O. Doebelin and D. N. Manik, Measurement Systems: Application and Design, 6th ed. New York, NY, USA: McGraw-Hill Education, 2011.

[2]. N. V. Raghavendra and L. Krishnamurthy, Engineering Metrology and Measurements, New Delhi, India: Oxford University Press, 2013.

Title	**Design, Manufacturing, and Thermal Power Lab**	Number	MEP3XX0
Department	Mechanical Engineering	L-T-P [C]	0-0-6 [3]
Offered for	B.Tech.	Type	PC
Prerequisite			

Part-I: Design of Machine Elements Lab

Objective: To impart hands-on skills in designing and analyzing machine elements using SolidWorks, focusing on static and fatigue loading conditions.
Learning Outcomes: Upon completion, students will proficiently design and model machine components for strength, fatigue, and efficiency using SolidWorks software. They will evaluate designs against failure criteria and generate detailed reports for real-world engineering applications.

Contents:
Lab 1: Review of SolidWorks: bolt and nut assembly.
Lab 2: Design of a cotter joint.
Lab 3: Design of flange coupling.
Lab 4: Design of a welded joint subjected to fatigue.
Lab 5: Design of a pin and bush coupling.
Lab 6: Design of a shaft.
Lab 7: Design of a spring subjected to fatigue.
Lab 8: Design a triple-riveted double-strap longitudinal butt joint and circumferential joint: boiler.
Lab 9: Design of connecting rod.
Lab 10: Design of screw jack.
Lab 11: Design and optimization of heat dissipation in the brake.

Text/Reference books

1. Shigley, J. E., Mischke, C., Nisbett, K., and Richard, B., (2008), Mechanical Engineering Design, McGraw Hill Education
2. Spotts, M. F. & Shamp, T. E., (2003), Design of Machine Elements, Prentice Hall
3. Sidheswar, K. N, & Sastry, P. V. V. V., (1997), Machine Drawing, Tata McGraw Hill
4. Cencil Jensen, Jay D. Helsel, Dennis R. Short, (2012), Engineering Drawing & Design, 7th Edition, TATA McGRAW-HILL
5. Mahadevan, K., Reddy, K.B. (2013), Design Data Handbook for Mechanical Engineers, CBS Publishers & Distributors

Online course material
1. https://help.solidworks.com/
2. https://www.ansys.com/en-in

Part-II: Machining and Automated Manufacturing Lab
Objective:

- To impart hands-on experience in conventional and CNC machining processes.
- To develop skills in CNC programming, simulation, and automated manufacturing systems.
- To enable analysis of machining quality through inspection and surface characterization.

Learning Outcomes:

- Apply machining processes for component fabrication.
- Develop and execute CNC programs for manufacturing.
- Evaluate machining quality and tool performance.
- Understand automation and advanced manufacturing technologies.

Experiments:

1. CNC Part programming and simulation for Milling operations
2. CNC part programming and simulation for Turning operations
3. Execution and verification of simulated CNC programs on machining and turning centers
4. Study and demonstration of an automated material handling system
5. Machining and fabrication of a specified component using Lathe operations
6. Machining operations on a drilling machine: Drilling, Counterboring, and Countersinking
7. Reconditioning of a single point cutting tools using tool grinding machine
8. Machining and finishing of a component using Milling and Grinding machines
9. Demonstration of a Laser-based manufacturing facility (Cutting, Welding and Marking)
10. Measurement and comparative analysis of surface roughness for Milling, Grinding, and Turning operations

Reference books

1. A. Overby, CNC Machining Handbook: Building, Programming, and Implementation, 1st ed. New York, NY, USA: McGraw-Hill, 2010.
2. P. Smid, CNC Handbook, 3rd ed. New York, NY, USA: McGraw-Hill, 2008.
3. E. Oberg, F. D. Jones, H. L. Horton, and H. H. Ryffel, Machinery's Handbook, 32nd ed. New York, NY, USA: Industrial Press, 2024.

Part-III: Power Systems Engineering Lab

Objective: The course aims to

1. To provide hands-on training on the operation of the different thermofluidic devices
2. To enable practical visualization of energy conversion and utilization processes
3. Introduce industry-oriented thermo-fluid process design problems.

Learning Outcomes: The students are expected to have the ability to:

1. Build thermo-fluid process models.
2. Ability to interpret data from experiments and usage of charts, software, and handbooks
3. To enable novel designs for new generation thermofluid technologies

Contents:

1. Calorific Value and Flame velocity measurement
2. 2S and 4S valve timing diagram
3. Heat Balance of IC Engine
4. Morse test
5. In cylinder diagnostics
6. Emission measurement
7. Gas turbine rig measurements
8. Gas turbine simulation
9. Thermal Power Plant with Steam Engine Trainer
10. Centrifugal Compressor Test Rig
11. Turbomachinery Compressor Design
12. Turbomachinery Turbine Design

Text/Reference books

1. Heywood, J.B. (1988). Internal Combustion Engine Fundamentals.
2. Ei-Wakil,M.M, (1985),Power Plant Technology, Tata McGraw-Hill
3. Saravanamuttoo, H. I. H., Rogers, G. F. C., Cohen, H., & Straznicky, P. V.. Gas turbine theory (6th ed.). Pearson Education Ltd

Title	**Automation, Control and Conditioning Lab**	Number	MEL3XX0
Department	Mechanical Engineering	L-T-P [C]	0–0–8 [4]
Offered for	B. Tech (ME)	Type	Program Compulsory (PC)
Prerequisite			

Part-I Control of Mechanical Systems Lab

Objectives

1. To develop an understanding of modelling, analysis, and design of control systems using both simulation tools and hardware experiments.
2. To apply theoretical concepts of control systems to real-world systems for analyzing performance, stability, and controller design.

Learning Outcomes

1. Students will be able to model, simulate, and analyze control systems, and design appropriate controllers using modern tools like MATLAB/Simulink.
2. Students will be able to implement and evaluate control strategies on real hardware systems and interpret system behavior under different operating conditions.

Contents

Simulation-Based Experiments (Using MATLAB/Simulink)

1. To perform a sensitivity analysis of a simple feedback control system using MATLAB/Simulink by studying the effect of parameter variations on system performance.
2. To model and simulate mechanical systems in MATLAB/Simulink by developing mathematical representations, such as transfer functions and state-space models.
3. To study the dynamic response and analyze the stability of a mechanical system by evaluating time-domain specifications and system poles.
4. To design a lead–lag compensator using the root locus method in order to improve the transient and steady-state performance of the system.
5. To design a controller using the frequency response method by analyzing Bode plots and ensuring desired gain and phase margins.
6. To design a controller using state-space techniques by implementing state feedback and achieving the desired system performance.

Hardware-Based Experiments

7. To study the filtering, system identification, and dynamic response of a DC motor by analyzing experimental data and reducing noise effects.
8. To implement P, PD, and lead–lag compensators for DC motor control and compare their effects on system performance.
9. To control and stabilize a rotary inverted pendulum system using appropriate feedback control techniques.
10. To study and control an active suspension system to improve stability and reduce the effect of external disturbances.
11. To control the liquid level in a coupled tank system by maintaining desired levels and analyzing the interaction between tanks.
12. To design and implement temperature control of a thermal system using feedback control methods.

Reference Books

1. Franklin, G. F., Powell, J. D. and Emami-Naeini, A. E. (2022), Feedback Control of Dynamic Systems, 8th Edition, Pearson India
2. Gopal, M. (2000), Control Systems - Principles & Design, 3rdEdition, Tata McGraw-Hill, 2015
3. Dorf, R.C. & Bishop, R.H. (2008), Modern Control Systems, 11th Edition, Pearson, 2009
4. Ogata, K. (2012), Modern Control Engineering, 5th Edition, Prentice-Hall India, 2012,
5. Nise, N.S. (2014), Control Systems Engineering, 6th Edition, John-Wiley India, 2014

Part-II: Design and Computational Mechanics Lab

Objective: The objective of this course is to provide a basic understanding of the critical regions in machine components subjected to multi-physical interactions and of the structure's topology optimization.

Learning Outcomes: After completion of this course, students will be able to:
1. Numerically solve differential equations using finite element methods.
2. Identify critical locations within the machine component.
3. Understand the mechanics of multi-physical interactions.

Contents:

1. Introduction to FEA solver: visualization of shape functions, steps for numerical solution of a boundary value problem, preprocessing, boundary conditions, and post-processing of the results (2 Exercise/4 Hours).
2. Generation of complex three-dimensional geometries in Gmsh/Ansys (1 Exercise/2 Hours).
3. Solution of the steady state heat equation and verification against analytical solution (1 Exercise/2 Hours).
4. Solution of two-dimensional (plane stress/strain) linear problems in solid mechanics, and verification against analytical solutions (1 Exercise/2 Hours).
5. Mesh refinement study and error estimation (1 Exercise/2 Hours).
6. Finding stress concentration factor for a plate with a circular hole subjected to far-field stresses (1 Exercise/2 Hours).
7. Solution of the transient heat equation and quantification of the effect of discretization and time increment on temperature solution (1 Exercise/2 Hours).
8. Solution of one-way thermo-elastic problems and the effect of geometric discontinuities (2 Exercises/4 Hours).
9. Topology optimization of rectangular beam and 3D printing (2 Exercises/4 Hours).
10. Topology optimization of the rectangular beam made up of functionally graded materials (1 Exercise/2 Hours).

Text/Reference books

1. Fish, Jacob, and Ted Belytschko. A first course in finite elements. Vol. 1. New York: Wiley, 2007.
2. Reddy, Junuthula Narasimha. Introduction to the finite element method. McGraw-Hill Education, 2019.
3. Bendsoe, Martin Philip, and Ole Sigmund. Topology optimization: theory, methods, and applications. Springer Science & Business Media, 2003.
4. Suresh, K., 2021. Design Optimization using MATLAB and SOLIDWORKS. Cambridge University Press.

Online course material

1. Ansys Academic Support | Tutorials & Training Materials
2. FEniCS documentation: fenicsproject.org.

Part-III: Thermofluids Lab

Objective: The course aims to

1. To provide hands-on training on the operation of the different thermofluidic devices
2. To enable practical visualization of energy conversion and utilization processes
3. Introduce industry-oriented thermo-fluid process design problems.

Learning Outcomes: The students are expected to have the ability to:
1. Build thermo-fluid process models.
2. Ability to interpret data from experiments and usage of charts, software, and handbooks
3. To enable novel designs for new generation thermofluid technologies

Contents

1. Vapor absorption refrigeration
2. Vapor compression refrigeration
3. Modeling and Analysis of VCRS
4. Ventilation experimentation facility and analysis
5. Modeling and Analysis of HVAC system

6. Steam jet refrigeration
7. Performance characteristics of Pelton turbine
8. Performance characteristics of Francis turbine
9. Performance characteristics of Centrifugal pump
10. Series & Parallel Centrifugal Pump Test Rig
11. Modeling pump network

Text/Reference books

1. Jordan, R. C. & Priester, G. B., (2003), Refrigeration & Air conditioning, Prentice Hall of India
2. Sheperd, D. G., (1983), Principles of Turbomachinery, Mac Millan

Part-IV: Smart Automation Lab

Objectives: The objectives of the course are to provide

1. hands-on exposure to mechatronic systems, industrial automation, and smart connected systems.
2. practical understanding of sensing, control, communication, monitoring, and integrated automation systems.

Learning Outcomes: The course will enable students to

1. Interface and integrate sensors, actuators, controllers, and industrial communication systems.
2. Develop basic supervisory, monitoring, and smart automation applications.

Course Contents

List of Experiments:

- Study and interfacing of industrial sensors for the measurement of quantities such as displacement, proximity, temperature, force, and speed (Experiment 1 and 2)
- Study of pneumatic, hydraulic, and stepper motor-based actuation systems for motion control (Experiment 3 and 4)
- Development of electro-pneumatic automation circuits for industrial applications (Experiment 5)
- PLC-HMI integration for process visualization and supervisory control (Experiment 6)
- Design and Development of a data acquisition system for manufacturing operations - Metal Casting and Machining (Experiment 7 and 8)
- Automated conveyor-based material handling and sorting systems (Experiment 9)
- Design of Machine Vision-Based Inspection (Experiment 10)
- Development of an integrated smart automation mini project involving sensing, control, communication, monitoring, and supervisory systems (Experiment 11 and 12)

References

1. Manuals for Equipments, Sensors and Data Acquisition in the Lab
2. Matlab Simulink for Mechatronic System Design, Mathworks
3. Factory Automation and Digitalization Solutions, Siemens AG

Title	**Basics of Finite Element Analysis**	Number	MEL4XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PE
Prerequisite			

Objective: (a) To introduce the fundamentals of finite element analysis for one-dimensional engineering problems and develop finite element formulations from governing differential equations, (b) To provide hands-on experience in MATLAB/Python-based FEM implementation and validation using commercial simulation software.

Learning Outcomes: Upon successful completion of the course, students will be able to formulate and solve one-dimensional structural, thermal, and vibration problems using finite element methods. Students will also be able to implement FEM algorithms computationally and validate numerical results with analytical and commercial software solutions.

Contents:

Introduction [3 Lectures]: Historical development and scope of FEM, General steps involved in FEM, Applications, and advantages. Comparison of FE and exact solutions, Matrix notation, treatment of boundary conditions, and solution for the unknown degree of freedom. Matlab implementation.

Finite element formulation from governing differential equation [8 Lectures]: Weighted residual method, Use of single continuous trial function. General weighted residual statement. Weak (variational) form of the weighted residual statement. Comparison of differential equation, weighted residual and weak forms. Piecewise continuous trial function solution of the weak form. Rayleigh-Ritz Method, collocation method, Least square method, Galerkin approach.

Finite Element Formulation for One-Dimensional Problems [10 Lectures]: Local approximation function and global approximation function, convergence, continuity, and completeness. Compatibility. Weak form for bar with varying cross-section, 1D linear and higher order shape functions, Discrete equation, Local and global stiffness matrix, force vector from weak form. Numerical integration, Gauss quadrature, transformation from physical space to parent space. Matlab implementation. Verification with analytical solutions. Patch test.

FE-formulation for Beam elements [5 Lectures]: Beam theory, governing differential equation and boundary conditions, Weak formulation, Hermite shape function, Derivation of local stiffness matrix, and Force vector. Matlab implementation. Verification with analytical solutions.

Finite element analysis of time-dependent problems [5 Lectures]: Transient heat conduction equation, Weak formulation, FE-formulation. Finite difference approximation for time. Explicit and implicit schemes. MATLAB/Python implementation of transient heat conduction problems.

Finite element analysis eigenvalue problems and structural dynamics [8 Lectures]: Introduction to eigenvalue problems in engineering applications, free vibration of a rod, free vibration of a beam. Structural dynamics, governing differential equations for a bar, including the inertia term. Weak formulation. FE-formulation. Consistent mass matrix and Lumped mass matrix. Explicit and Implicit time integration scheme. Newmark's method. MATLAB/Python implementation of vibration problems.

Text/Reference books

1. Bathe, K.J., 2006. Finite element procedures. Klaus-Jurgen Bathe.
2. Fish, J. and Belytschko, T., 2007. A first course in finite elements (Vol. 1). New York: Wiley.
3. Reddy, J.N., 2026. An introduction to the finite element method. In Dynamics of Earth's Fluid System (pp. 199-226). CRC Press.
4. Hughes, T.J., 2003. The finite element method: linear static and dynamic finite element analysis. Courier Corporation.
5. Hutton, D., 2004. Fundamentals of finite element analysis. McGraw-Hill.
6. Cook, R.D., 2007. Concepts and applications of finite element analysis. John wiley & sons.
7. Seshu, P., 2003. Textbook of finite element analysis. PHI Learning Pvt. Ltd.

Online course material

1. https://nptel.ac.in/courses/112106135
2. https://onlinecourses.nptel.ac.in/noc20_me60/preview

Title	**Mechanical Vibrations**	Number	MEL4XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PE
Prerequisite			

Objective: The objective of this course is to introduce fundamental principles of vibration analysis in SDOF, MDOF, and continuous systems, along with understanding vibration response, resonance, modal behavior, and control techniques.

Learning Outcomes: Upon completion, students will be able to formulate equations of motion, analyze free and forced vibrations, determine natural frequencies and mode shapes, and apply modal analysis and vibration control methods to engineering problems.

Contents:

Single degree of freedom (SDOF) systems [14 Lectures]*:* Introduction, Basic concepts of vibrations, Elementary parts of vibrating systems, Equation of motion of SDOF system, Undamped free and forced vibrations of SDOF system, Damped free and forced vibrations of SDOF system, Frequency response, resonance, Beating phenomenon, Vibration under general forcing condition, Duhamel's integral, Impulse loading, whirling of shaft and critical speed.

Multi-degree of freedom (MDOF) systems [12 Lectures] : Two degree of freedom systems, Multi degree of freedom systems, Equation of motion, Free and forced vibration analysis, Normal modes of vibration, natural frequencies and mode shapes, Modal superposition theorem, Modal analysis.

Continuous systems [11 Lectures]: Transverse vibration of a string, Axial and torsional vibration in rod, Flexural vibration in beam, Rayleigh's method, Rayleigh-Ritz method.

Vibration control [2 Lectures]: Vibration isolation, vibration absorber.

Text/Reference books

1. Thompson, W.T., (1993), Theory of Vibration with Applications, Prentice Hall
2. Rao, S.S., (1990), Mechanical Vibrations, Prentice Hall.
3. Meirovitch, L., (1986), Elements of vibration analysis,, Prentice Hall
4. Weaver Jr, W., Timoshenko, S.P. and Young, D.H., 1991. Vibration problems in engineering. John Wiley & Sons.

Online course material
https://nptel.ac.in/courses/101105081
https://nptel.ac.in/courses/112103112

Title	**Introduction to Mechatronics**	Number	MEL4XXX
Department	Mechanical Engineering	L-T-P[C]	3–0–0 [3]
Offered for	B.Tech.	Type	Program Elective
Prerequisite			

Objectives: The objectives of the course are -

1. Develop an understanding of the interdisciplinary principles of mechatronics, automation, and robotic systems.
2. Understand the working principles of sensors, actuators, embedded systems, and control systems used in automation applications.
3. Apply system integration concepts to design and implement basic mechatronic and automation solutions.

Learning Outcomes: Upon completion of the course, students will be able to achieve the following:

1. Explain the architecture and operation of mechatronic and industrial automation systems.
2. Integrate sensors and actuators using embedded systems or PLC-based platforms.
3. Design and implement simple automation or robotic systems for engineering applications.

Course Content

Introduction: Evolution of automation; Industrial revolutions; Components of mechatronic systems; Applications in manufacturing and robotic process automation [3 lectures]

Mechanical Systems and motion elements: Mechanical components and mechanisms, Power transmission systems, Linear motion systems, and machine structures [4 lectures]

Actuators and Drive Systems: Classification of actuators; Electrical actuators- DC motors, Servo motors, Stepper motors; Pneumatic actuators, Hydraulic actuators, Motor drivers and motion transmission systems, Speed and position control concepts, Applications in robotics and industrial automation [10 lectures]

Sensors and Industrial Instrumentation: Fundamentals and characteristics of sensors, Position, proximity, and motion sensors, Force, pressure, and temperature sensors, Signal conditioning and data acquisition, Sensor interfacing with embedded systems, Applications in robotics and industrial automation [8 lectures]

Embedded Systems and Electronics for Automation: Basic electronics for automation systems, Analog and digital signals, Sensor and actuator interfacing, ADC, DAC, and PWM concepts, Communication protocols, Embedded programming basics [8 lectures]

PLCs and Industrial Automation Case Studies: PLC architecture and input/output modules, Ladder logic programming fundamentals, Timers, counters, and sequencing operations, HMI and SCADA basics, Industrial automation case studies (Conveyor automation, Automated sorting systems, PLC-based filling and packaging systems) [6 Lectures]

Reference Books

1. Alciatore D. G. and Histand M. B., *Introduction to Mechatronics and Measurement Systems*. McGraw-Hill Education.
2. Shetty D. and Kolk R., *Mechatronics System Design*. Cengage Learning.

Self-Learning Material

1. Prof. Prasanna Gandhi, “Design of Mechatronic Systems” at Indian Institute of Technology Bombay. https://onlinecourses.nptel.ac.in/e-learning/preview/noc23_me136

Title	**Introduction to Robotics**	Number	MEL4XXX
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech.	Type	PE
Prerequisite			

Objectives

1. To introduce fundamental aspects of modelling and control of robot manipulators.
2. To provide a brief of results from geometry, kinematics, dynamics, motion planning and control

Learning Outcomes

1. The course will equip students with theoretical and practical knowledge of robot modelling, programming and control.

Contents (This course is subset of Robotics offered for M.Tech. in AMD and includes 3 fractals)

Fractal I: Robot Modeling

Position Kinematics: Transformations, Rigid Motions, Forward Kinematics: Denavit-Hartenberg Convention, Inverse Kinematics [6 lectures]

Velocity Kinematics – The Jacobian: Angular and Linear Velocity: Singularities, Accelerations [4 lectures]

Robot Dynamics: Equations of Motion, Kinetic and Potential Energy, Euler-Lagrange Equations, Recursive Newton-Euler Formulation [4 lectures]

Fractal II: Robot Motion Planning and Visual Servoing

Path and Trajectory Planning: Path vs. Trajectory, The Configuration Space, Path Planning in Configuration Space, Probabilistic Roadmap Planner, Potential Fields, RRTs, Trajectory Planning, Point To Point Motion, Paths Specified by Via Points, [6 lectures]

Visual Servoing: Computer Vision for Visual Servoing, The Geometry of Image Formation, Camera Calibration, Camera Motion and Interaction Matrix, Image-Based Control Laws, Relation between End Effector and Camera Motions, Partitioned Approaches [6 lectures]

Fractal III: **Robot Control**

Linear Control: Feedback and closed-loop control, second-order linear systems, control of second-order systems, Control-law partitioning, trajectory-following control, disturbance rejection, Continuous vs. Discrete time control, modelling and control of a single joint, architecture of an industrial-robot controller [5 lectures]

Nonlinear control: Nonlinear and time-varying systems, multi-input, multi-output control systems, control of manipulators, practical considerations, Lyapunov stability analysis, cartesian-based control systems, adaptive control [5 lectures]

Force Control: Interaction with Environment, Force Control, Hybrid Force/Motion control [3 lectures]

Reference Books

1. Saha S. K., Introduction to robotics. Tata McGraw-Hill Education
2. Spong M. W., Hutchinson, S., and Vidyasagar, M., Robot modeling and control. New York: Wiley.
3. Craig J. J, Introduction to robotics: mechanics and control. Pearson/Prentice Hall

Self Learning Material

1. https://see.stanford.edu/Course/CS223A

Title	**Planning and Decision Making for Robots**	Number	MEL7XXX
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D.	Type	Program Elective
Prerequisite	Introduction to Robotics or Autonomous Systems		

Objectives

Provide the fundamental understanding of concepts like Motion Planning, State Estimation, Reinforcement learning, and Optimization-based control, etc.

Learning Outcomes

The students are expected to design and implement algorithms for planning and decision-making for robots.

Contents:

Fundamentals of probability, random processes, and optimization [6 Lectures]

Optimization-based Control: Collocation, Shooting, MPC, Contact-Invariant Optimization [5 Lectures]

Markov Decision Processes: Exact Methods, Discretization of Continuous State Space MDPs [4 Lectures]

Motion Planning: Trajectory optimization, 3D poses [2 lectures]

Kalman Filtering, Extended Kalman Filter, Unscented Kalman Filter, Smoother, MAP, Maximum Likelihood, EM, KF parameter estimation, Particle Filters [6 Lectures]

Introduction to Reinforcement Learning and Deep RL [3 Lectures]

Value-based methods for deep RL, Policy gradient methods for deep RL [4 Lectures]

Model-based methods for deep RL, Deep reinforcement learning beyond MDPs [4 Lectures]

Deep RL for Motion Planning [5 Lectures]

Reference Books:

1. Bongard, Josh. "Probabilistic Robotics. Sebastian Thrun, Wolfram Burgard, and Dieter Fox.(2005, MIT Press.) 647 pages." (2008): 227-229.
2. Koller, Daphne, and Nir Friedman. Probabilistic graphical models: principles and techniques. MIT press, 2009.
3. Boyd, Stephen, Stephen P. Boyd, and Lieven Vandenberghe. Convex optimization. Cambridge university press, 2004.
4. Sutton, R. S., & Barto, A. G. (2018). Reinforcement learning: An introduction. MIT press.

Online Course Material:

1. Prof. Pieter Abbeel, “Advanced Robotics” at University of California at Berkeley. https://people.eecs.berkeley.edu/~pabbeel/cs287-fa19/

Title	**Mobile Robots**	Number	MELXXXX
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech., M.Tech., Ph.D.	Type	Program Elective
Prerequisite	Introduction to Robotics or Autonomous Systems		

Objectives

1. Introduce basic concepts of modelling of various mobile robots and their applications.
2. Provide an understanding of the navigation and control of mobile robots in various practical applications.

Learning Outcomes

3. The students are expected to have the ability to model and design stabilizing control algorithms for various mobile robots operating individually or collectively.

Contents:

Introduction to Mobile Robotics [13]
Kinematics of wheeled mobile robots, Forward and Inverse Kinematics [4]
Motion constraints [2]
Nonholonomic mobile robots [4]
Dynamic model of differential drive robot [3]

Control of Wheeled Mobile Systems [15]
Orientation, forward-motion, and reference position control [3]
Trajectory tracking control [3]
Feedback linearization [2]
Lyapunov-based control design [3]
Collective coordination and control [4]

Mobile Robot Sensors, and Path Planning [11]
Brief introduction to mobile robot sensors [3]
Basic principles of navigation [2]
Model-based robot path planning [3]
Introduction to mobile robot localization and mapping [3]

Reference Books:

1. Klancar, G., Zdesar, A., Blazic, S., & Skrjanc, I. (2017). Wheeled mobile robotics: From fundamentals towards autonomous systems. Butterworth-Heinemann.
2. Tzafestas, S. G. (2013). Introduction to mobile robot control. Elsevier.
3. Cook, G., & Zhang, F. (2020). Mobile Robots: Navigation, Control and Sensing, Surface Robots and AUVs. John Wiley & Sons.

Online Course Material:

1. https://www.coursera.org/learn/mobile-robot

Title	**Micro-nano Flows**	Number	MEL4XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech (ME)	Type	PE
Prerequisite	Fluid Mechanics & Heat Transfer		

Objective:

1. To inculcate the physics of liquid transport in micro-nano fabricated systems
2. To introduce theory and physical principles of fluid mechanics on the microscale

Learning Outcomes: The students are expected to have the ability to:

1. Understand physics involved in fluid flow in micro-nano fabricated systems
2. Solve problems of fluid flows in microchannels

Contents:

Hydrodynamics review (3 Lectures): Continuum assumption, Conservation equations at Continuum level, Low Reynolds's number flows, Creeping flows, Physics of miniaturization, Scaling laws

Fluid mechanics at micro-scale (4 Lectures): Intermolecular forces, States of matter, Governing equations, Constitutive relations, Gas and liquid flows, Boundary conditions, Slip theory-first order and higher order models, Transition to turbulence

Shear-driven flows (5 Lectures): Couette flow slip flow regime, Transition and free molecular regime, Oscillatory Couette flow, Cavity flow

Pressure-driven flows (6 Lectures): Micro channel flows, Slip flow regime, Transition and free molecular regime, Entrance flows, Effects of roughness, Compressible micro channel flows

Thermal Effects at the microscale (5 Lectures): Thermal creep, Heat transfer in Poiseuille microflows, pressure-driven microflows, and force-driven microflows, Heat transfer in Couette microflows

Surface tension-driven flows (5 Lectures): Surface tension and interfacial energy, Different forms of Young's equation, Thin films, Capillary flows

Electrokinetic flows (5 Lectures): Electrokinetic effects, Electric double layer, Governing equations, Electroosmotic flows, Electrophoresis, Dielectrophoresis, Electrocapillary, Continuous electro-wetting, Direct electro-wetting, Electro-wetting on dielectric

Microfluidic devices (6 Lectures): Micropumps, Microvalves, Microflow sensors, Micromixers, Micropropulsion and Micronozzle flows

Reference books

1. G. M. Karniadakis, A. Beskok, N. Aluru, Microflows and Nanoflows: Fundamentals and Simulation, Springer, 2005
2. N. T. Nguyen and S. T. Wereley, Fundamentals and Applications of Microfludics, Artech House Inc., 2002
3. P. Tabeling, Introduction to Microfluidics, Oxford University Press, 2005

Online course material

1. https://nptel.ac.in/courses/112105187/
2. https://nptel.ac.in/courses/112106169/

Title	**Computational Fluid Dynamics and Heat Transfer**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. / M.Tech. (Program Elective) - Mechanical / Aerospace	Type	PE
Prerequisite	--		

Objective:

1. To develop a strong foundation in the governing equations, numerical methods, and physical modeling of fluid flow and heat transfer.
2. To apply CFD techniques using software tools, with an understanding of discretization, convergence, and solution accuracy.
3. To analyze and interpret CFD results for real engineering problems and practical modeling considerations.

Learning Outcomes:

1. Demonstrate understanding of the key theories, approaches, and methodologies underpinning CFD.
2. Apply standard CFD procedures using modern software tools.
3. Utilize CFD analysis to solve and interpret real-world engineering design problems.

Contents:

Introduction to CFD [6 Lectures]*:* CFD workflow, Applications of CFD, Review of governing equations for heat transfer and fluid flow; Brief introduction of ODE (IVP and BVP) and PDE, Initial and Boundary conditions, classification of PDE, various methods to solve PDE numerically

Finite Difference Method [12 Lectures]: Taylor series expansion; derivation of finite difference schemes; accuracy and order of discretization; stability and truncation error analysis of finite difference formulations, Application to heat transfer problems, Steady state heat conduction with heat source, Convective- diffusive system – Upwind scheme, Transient heat conduction: explicit, implicit and Crank – Nicolson methods, Handling variable thermo-physical properties, fluid flow problems, Solution of simultaneous equations: direct and iterative methods; Jacobi and various Gauss-Seidel methods, Gauss-elimination, TDMA (Thomas), Gauss-Jordan, other direct and indirect methods.

Finite Volume Method [12 Lectures]: Finite Volume Method formulation, calculation of fluxes at faces of control volume, Source term linearization Discretization for convective terms: upwind, power law, QUICK schemes, staggered grid and solution of incompressible fluid flow, SIMPLE algorithm and extensions to SIMPLER, SIMPLEC and PISO, multigrid techniques

Advanced Topics [9 Lectures]: Solution convergence, Verification and Validation, Boundary conditions, Grid generation, Turbulent flows; Reynolds-averaged Navier–Stokes (RANS) equations; Boussinesq approximation; RANS models; Introduction to high-fidelity simulations (LES/DNS), Data driven techniques

Text/Reference books

1. Versteeg, H.K, and Malalasekera, W., Computational fluid Dynamics-The finite volume method, 2007
2. Anderson, John, D. Computational Fluid Dynamics: The Basics with Applications, 1995
3. Patankar, S. Numerical Heat Transfer and Fluid Flow, 1980

Online course material

https://nptel.ac.in/courses/112105045
https://www.fluidmechanics101.com/

Title	**Conduction and Radiation**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech., M.Tech., and Ph.D.	Type	PE
Prerequisite			

Objective:

1. To provide a deeper and more thorough analytical background for 1-D, 2-D, 3-D steady and transient multi-scale heat conduction problems relevant to multi-disciplinary engineering applications.
2. Exposure to the concepts of solving different classes of partial differential equations (PDE) associated with heat conduction problems analytically and numerically.
3. To provide a deeper and more thorough analytical background for radiation heat transfer associated with opaque surfaces and participating media

Learning Outcomes:

1. Enhanced analytical ability to solve PDEs associated with multidimensional conduction problems.
2. Ability to identify types of boundary conditions.
3. Ability to produce codes to solve different classes of conduction problems and familiarity with analytical and empirical radiation models used in CFD tools.

Contents:

Conduction:

1. Recapitulation, 3-D conduction, isotropic, orthotropic and anisotropic solids, Mathematical formulation, analytical solutions, variation of parameters **(8 Lectures)**
2. Integral method, periodic boundary conditions, Duhamel's theorem and Green's function, Stationary and moving heat sources and sinks, Moving boundary problems **(8 Lectures)**
3. Inverse heat conduction problems, Micro-scale heat transfer, Bio-heat transfer, combined-mode heat transfer, integro-differential equations, introduction to Monte-Carlo technique **(8 Lectures)**

Radiation:

4. Recapitulation, Radiative properties of opaque surfaces, Configuration factors, **(5 Lectures)**
5. Enclosure analysis for diffuse-gray surfaces and non-diffuse non-gray surfaces, Radiation in absorbing, emitting and scattering media **(5 Lectures)**
6. Engineering treatment of gas radiation in enclosures, Radiation measurements **(5 Lectures)**

Reference Books:

1. Ozisik, Necati M., Heat conduction, John Wiley & Sons.
2. Howell, J. R., Menguc M. P., and Robert Siegel. Thermal radiation heat transfer. CRC Press.

Title	**Convective Heat and Mass Transfer**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech., M.Tech., and Ph.D.	Type	PE
Prerequisite			

Objective:

1. To provide a deeper and more thorough analytical and numerical background for coupled heat and fluid flow problems relevant to multi-disciplinary engineering applications.
2. Exposure to the concept of thermal and hydrodynamic boundary layers and their interactions.
3. Exposure to solution techniques of solving system of coupled non-linear PDEs using analytical and numerical methods.

Learning Outcomes:

1. Enhanced analytical and numerical ability to solve system of coupled non-linear PDEs associated with multidimensional convection problems.
2. Ability to Identify types of boundary conditions and coupling constraints.
3. Ability to produce codes to solve different classes of coupled heat and fluid flow problems.

Contents:

1. Conservation equations for mass, momentum, energy and species. Boundary layer flows, similarity parameters, dimensionless numbers. (**9 Lectures**)
2. External laminar flow heat transfer. Internal laminar flow heat transfer, entrance region.**(9 Lectures**)
3. Turbulent flow heat transfer, turbulent Prandtl number, external and internal flows. **(6 Lectures)**
4. Natural convection in external and bounded flows. Rayleigh-Bénard Convection **(6 Lectures)**
5. Mixed convection. Boiling – pool boiling and forced convection boiling in tubes. Condensation over a plate, tube and tube banks. Mass transfer. Applications to engineering problems. **(9 Lectures)**

Reference Books:

1. Bejan, A., Convection heat transfer. John Wiley
2. Kakaç, S., Shah R. K., and Aung, W. Handbook of single-phase convective heat transfer
3. Kays, W. M., Convective heat and mass transfer, Tata McGraw-Hill

Title	**Numerical Methods for Mechanical Engineers**	Number	MEL4XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech.	Type	PE
Prerequisite	UG Mathematics, Basic computer programming		

Objective:

1. To introduce the student to basic numerical analysis techniques which are used to solve science and engineering problems.
2. To provide students with experience in programming some of these techniques to help them gain familiarity with the methods and the errors associated with them.

Learning Outcomes:

1. Preparing industry-ready graduates equipped with computational skills required in FVM, FEM, MD etc.
2. Ability to analyse results, along with code verification and validation.

Contents:

1. **Introduction [2 lectures]**: Algorithms, convergence, representation of numbers with finite bits and associated errors.
2. **Root finding [6 lectures]:** Methods of bisection, false position, fixed-point iteration, Newton-Raphson method, secant method, Brent's method
3. **Linear Algebra [8 lectures]:** Review of linear algebra – Matrix operations; Reduced row echelon form and Rank, least square approximation, Eigenvalues and Eigenvectors; Singular Value Decomposition (SVD); condition number, LU decomposition, direct factorization, iterative techniques, conjugate gradient methods, nonlinear systems of equations.
4. **Optimization [7 lectures]:** Golden section search method, Parabolic interpolation, Newton's Method, Brent's method; Multidimensional unconstrained – Direct methods, gradient methods; Constrained Optimization: Linear Programming, nonlinear constrained optimization, Lagrange multipliers
5. **Curve fitting [4 lecture]:** Interpolation - Lagrange polynomial, piecewise linear, spline interpolation; Least Squares Regression
6. **Differentiation and Integration [6 lectures]:** Taylor series-based differentiation schemes, Richardson extrapolation, Trapezoidal, Simpson's, and Quadratures
7. **Ordinary Differential Equations [6 lectures]:** Initial value problem: Euler's method, Runge-Kutta methods, Convergence and stability, Stiffness; Boundary value problem: Finite difference, shooting method, predictor-corrector method; Inverse problems.

Textbook:

1. Bradie, B. (2006). A friendly introduction to numerical analysis. Pearson Education India.
2. Chapra, S. C., & Canale, R. P. (2011). Numerical methods for engineers (Vol. 1221). New York: McGraw-Hill.

Reference Books:

3. Press, W. H., Teukolsky, S. A., Vetterling, W. T., & Flannery, B. P. (2007). Numerical recipes in C++. The art of scientific computing, 2, 1002.
4. Chung, T. (2002). Computational Fluid Dynamics. Cambridge: Cambridge University Press. doi:10.1017/CBO9780511606205
5. Golub, G.H. and Van Loan, C.F., (1996). Matrix computations, 3rd edition, The Johns Hopkins University Press. Baltimore, MD

Online Materials:

6. Dr. Ameeya Kumar Nayak and Dr. Sanjeev Kumar, Numerical Methods, NPTEL Course Material, Dept of Mathematics, IIT Roorkee; URL: https://archive.nptel.ac.in/courses/111/107/111107105/
7. Prof. Boo Cheong Khoo, Prof. Jacob White, Prof. Jaime Peraire, Prof. Anthony T. Patera, Numerical Methods for Partial Differential Equations, OpenCourseWare, MIT, URL: https://ocw.mit.edu/courses/16-920j-numerical-methods-for-partial-differential-equations-sma-5212-spring-2003/pages/lecture-notes/

Title	**Fundamentals of Metal Forming**	Number	MEL7XXX
Department	Mechanical Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech.	Type	PE
Prerequisite	Strength of Material, Manufacturing Process		

Objectives

The instructor will:

1. To develop fundamental understanding of stress, strain, plasticity, and material behavior relevant to metal forming processes.
2. To analyze sheet metal deformation mechanisms, yielding behavior, formability, and instability during various sheet forming operations.
3. To study the mechanics and applications of bending, deep drawing, stamping, and hydroforming processes used in industrial sheet metal manufacturing.

Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Apply concepts of stress–strain analysis, plastic deformation, and material properties to metal forming problems.
2. Analyze sheet metal forming processes considering yielding, work hardening, formability, deformation modes, and failure mechanisms.
3. Evaluate and design sheet metal operations such as bending, stamping, deep drawing, and hydroforming with consideration of springback, instability, and forming limits.

Contents:

Introduction and Fundamentals of Metal Forming Mechanics **[11 Lectures]**: Introduction to the course: different metal forming processes, importance of plasticity in the course. Review-Analysis of stress: transformation relations, principal stresses and directions, maximum normal and shear stresses, invariants, hydrostatic and deviatoric parts; Analysis of (infinitesimal) strain: transformation relations, principal strains, invariants, hydrostatic and deviatoric parts; (Infinitesimal) rotation, Stress-strain relations for isotropic, linearly elastic material. Description of Material properties: Introduction, Tensile test, Effect of properties on forming and formability.

Sheet Metal Forming and Plastic Deformation Analysis **[14 Lectures]**: Sheet deformation processes: Uni-axial tension, general sheet forming processes, yielding, flow rule, work hardening hypothesis, work of plastic deformation, isotropic yield functions, isotropic yield functions, effective stress and strain, Sheet deformation in plane stress, strain distributions, strain diagram, deformation modes, effective stress-strain laws, principal tensions, Stamping analysis: 2-D modeling of stamping, stretch and draw ratios in stamping.

Load instability and tearing: uniaxial tension of perfect strip and imperfect strip, tensile instability in stretching continuous sheet, tensile instability in stretching continuous sheet.

Sheet Metal Formability, Bending and Deep Drawing **[14 Lectures]**: Sheet formability: Forming limit curve - concept and evaluation, FLC, formability tests, theoretical prediction, factors affecting FLC, Sheet bending: Variables in bending a continuous sheet, equilibrium conditions, material models, bending without tension, springback prediction Cylindrical deep drawing & Stretching circular shells: Drawing the flange, cup height, redrawing cylindrical cups, wall ironing of deep drawn cups, Analysis of bulging with fluid pressure, stretching over punch, Hydroforming, Demonstration on sheet forming operations - stretching, bending, deep drawing.

Textbooks

[1] R. H. Wagoner, J. L. Chenot, Fundamentals of metal forming, John Wiley and Sons, 1997.
[2] D. Banabic, Sheet metal forming processes Constitutive modeling and numerical simulation, Springer-Verlag Berlin Heidelberg, 2010

Reference Books

1. Z. Marciniak, J. L. Duncan, S. J. Hu, Mechanics of sheet metal forming, Elsevier, Butterworth-Heinemann, 2002
2. W. F. Hosford, R. M. Caddell, Metal forming Mechanics and Metallurgy, Printice-Hall, 2007
3. J. Chakrabarty, Theory of Plasticity, McGraw Hill, 1998.
4. D. W. A. Rees, Basic engineering plasticity, Elsevier, 2006

Online Course Material

https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_me90

Title	**Micro and nano Manufacturing**	Number	MEL4XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PE
Prerequisite	--		

Objectives:
1. Provide exposure to students with emerging manufacturing techniques for producing micro- and nano level products

Learning Outcomes:
The students will have the ability to:
1. Able to appreciate the processes employed for the fabrication of miniaturized products.
2. Explain the physical description, modelling and experimentations for manufacturing at micro and nano scale.

Contents:

An overview of micro and nano mechanical systems and their applications in Mechanical Engineering [5 Lectures]: Nanomachines, Nanomechanics, Nano-metrology, Nano-tribology, Nano-fluidics, Nano-manufacturing.
MEMS Micro fabrication [7 Lectures]: Science, modelling and experimental aspect of diffusion, thermal oxidation, ion implantation, rapid thermal processing, optical lithography, photoresists, non-optical photolithography, vacuum processes and plasmas, and etching processes, thin film manufacturing using physical vapor deposition, chemical vapor deposition, epitaxial growth processes.
Silicon Micromachining [5 Lectures]: anisotropic wet chemical etching, wafer bonding, dry plasma etching, surface micromachining.
Mechanical Micromachining [6 Lectures]: Abrasive micro grinding, micro milling, micro electro discharge machining, micro electro chemical machining, nano grinding.
Beam method [2 Lectures]: focused ion beam and electron beam machining.
Fabrication of nano materials and nano crystalline materials [6 Lectures]: Characterization techniques such as SEM, SPM, AFM, TEM.
Micro-nano patterned surfaces for functional devices [6 Lectures]: Application potential of micro and nano structured surfaces.
Recent advances in micro & nanofabrication [2 lectures]

Text/Reference books:
Text Book:
1. Drexler K E, Nanosystem: Molecular Machinery, Manufacturing and Computation, John Wiley
2. Rai-Choudhuri P, Handbook of Microlithography, Micromachining and Microfabrication, SPIE Press
3. McGeough J A, Micromachining of Engineering Materials, Marcel Dekkar

Reference Books:
1. Jain V K,(2011), Introduction to micromachining, Narosa Publications
2. Cabrini S and Hawata S, (2012), Nano fabrication hand book – CRC press

Self-Learning Material:
K N Bhat, Micro- and Smart Systems, NPTEL Course Material Department of Electrical Engineering, IISc Bangalore, https://nptel.ac.in/courses/112108092/

Title	Advanced Machining	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech./ M. Tech.	Type	Program Elective
Prerequisite			

Objectives

The course aims at imparting knowledge related to:

1. Understanding the principles, advancements, and applications of advanced manufacturing techniques, including material removal mechanisms, process capabilities, and recent innovations.
2. Gaining insights into modern manufacturing practices involving automation, additive manufacturing, AI-driven design, and hybrid manufacturing approaches.

Learning Outcomes

The students are expected to:

1. Develop the ability to analyse and apply manufacturing techniques, including both conventional and advanced processes.
2. Understand Industry 4.0 practices to optimize manufacturing efficiency and innovation.

Contents:

1. Introduction: Importance of manufacturing, introduction to advanced manufacturing processes, historical perspective, classification and overview of non-conventional machining (4 Lectures).

2. Advanced manufacturing processes (5 Lectures): classification, overview of working principles, material removal mechanisms, process capabilities, applications and limitations

3. Mechanical Material removal – Abrasive jet Machining, Abrasive water jet machining, Ultrasonic Machining – Basic Mechanism, process parameters, applications and defects/limitations. (8 Lectures)

4. Thermal Material removal – Electro-discharge machining, Electron beam machining, Laser beam machining, Ion Beam Machining – Basic Mechanism, process parameters, applications and defects/limitations. (10 Lectures)

5. Electro chemical machining: Basic principles of electrochemical machining, Mechanism of material removal and estimation of MRR, Electrochemical grinding and Electro-stream drilling, Electrochemical turning, Electrochemical Milling, Electrochemical deburring, Electrochemical boring and Allied processes – Basic Mechanism, process parameters, applications and defects/limitations. (8 Lectures)

6. Current trends in advanced manufacturing practices (4 Lectures): Introduction to Industry 4.0, Industrial IoT, sustainable manufacturing, ML/AI applications in advanced manufacturing, Industrial case studies.

Reference Books

1. Advanced Machining Processes, V. K. Jain, Allied Publishers
2. Fundamental of modern manufacturing Materials, Processes and systems, M.P. Groover
3. Manufacturing Science, Ghosh A., and Mallick A. K., EWP
4. Advanced Machining Processes, Hasan El-Hofy, McGraw-Hill

Online Course Material:

1. https://nptel.ac.in/courses/112104028
2. https://nptel.ac.in/courses/112104204
3. https://nptel.ac.in/courses/112107144/#
4. https://nptel.ac.in/courses/112105127/

Title	**Computer-Aided Manufacturing**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. (ME)	Type	PE
Prerequisite	--		

Objectives: The Instructor will:

- provide understanding of CNC Programming, Machine Tools and strategies for manufacturing freeform geometries using CAD/CAM systems
- introduce concepts of computer-aided inspection using CMM and machine vision systems
- enable understanding of computer-aided process planning and integration of CAD–CAM–CAI systems

Learning Outcomes: The students are expected to

- develop advanced CNC part programming and understand machining strategies for freeform geometries using CAD/CAM systems
- appreciate principles of computer-aided inspection using CMM and machine vision systems
- understand integration of CAD, CAM, CAI, and CAPP in digital manufacturing environments

Contents:

Introduction to CNC Systems: Review of CNC systems and architecture, Open architecture CNC systems, Direct and Distributed Numerical Control (DNC), Flexible Manufacturing Systems (FMS), Computer Integrated Manufacturing (CIM), CNC communication interfaces and industrial networking concepts (4 Lectures)

CNC Hardware: Mechanical structure of CNC machine tools, Feed drive systems and spindle systems, Servo motors and drives, Encoders and feedback systems, Ball screws and guideways, CNC interpolators and trajectory generation, PLC integration in CNC systems, Sensors in CNC machines, Sources of geometric, thermal, and dynamic errors in CNC machine tools (6 Lectures)

CNC Programming: Parametric and macro programming, Subroutines and advanced canned cycles, Tool length and cutter radius compensation, Polar and cylindrical interpolation, Multi-axis machining fundamentals, High-speed machining concepts, STEP-NC overview, Controller-specific programming considerations for turning and milling operations (7 Lectures)

Machining of Freeform Surfaces: Mathematical representation of curves and surfaces, Bézier curves, B-splines and NURBS, Surface representation methods for manufacturing, Toolpath generation for sculptured surfaces, Toolpath strategies for freeform machining, 3-axis and 5-axis machining concepts, Gouge avoidance and collision detection, Surface finish and chordal deviation, CAM simulation and verification (9 Lectures)

CAD/CAM Integration: CAD data exchange formats including IGES, STEP, and STL, Feature-based modeling, Associative manufacturing concepts, CAM workflow and post-processing, Virtual machining and CNC program verification (3 Lectures)

Computer Aided Inspection: Fundamentals of metrology in digital manufacturing, Coordinate Measuring Machines (CMM), Contact and non-contact inspection methods, CMM probing systems and programming, CAD-based inspection planning, Introduction to Geometric Dimensioning and Tolerancing (GD&T), Machine vision systems, Image acquisition and processing basics, Vision-based dimensional inspection, Integration of inspection systems with CAD/CAM environments (5 Lectures)

Computer Aided Process Planning (CAPP): Variant and generative CAPP, Feature recognition methods, Setup and operation sequencing, Knowledge-based process planning, AI-assisted process planning concepts, Integration of CAD–CAPP–CAM–CAI systems (5 Lectures)

Reference Books:

1) Smid P, CNC Programming Handbook, Industrial Press, 2008
2) Madison J, CNC Machining Handbook: Basic Theory, Production Data and Machining Procedures, Industrial Press, 1996
3) Marciniak K, Geometric Modeling for Numerically Controlled Machining, Oxford University Press, 1991
4) Mortenson M, Geometric Modeling, John Wiley and Sons, 1985

Online Course Material:

- NPTEL Lecture Course on Computer Aided Design and Manufacturing by Prof. Anoop Chawla and Prof. P V Madhusudhan Rao, https://nptel.ac.in/courses/112102101

Title	**Casting and Allied Processes**	Number	MEL4XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PE
Prerequisite	--		

Objectives:
The Instructor will:
1. Impart fundamental understanding of various casting processes.
2. Discuss fundamentals of solidification, design principles, melting techniques etc.

Learning Outcomes:
The students will have the ability to:
1. Appreciate various casting processes and its selection for different materials
2. Design systems and components necessary for obtaining cast components.

Contents:
Casting Processes [8 Lectures]: classification, characteristics of sand casting processes, metal mould casting processes and casting processes using other mould/core materials, Pattern materials, types of patterns, Mould and core making materials and their characteristics.
Technology of Selected Casting Processes [6 Lectures]: Clay bonded, oil bonded, synthetic resin bonded, and inorganic material bonded mould and core making processes. Sand additives and mould coatings. Metal mould casting processes, centrifugal and continuous casting processes.
Solidification [7 Lectures]: Nucleation and grain growth, Solidification of pure metals, short and long freezing range alloys. Rate of solidification, Solidification contraction.
Microstructure study and testing [5 Lecture]: Macrostructure and Microstructure Fluidity and its measurement. Mould-metal interface reactions.
Design of the casting system [7 Lecture]: Gating and Risering design & analysis.
Melting and quality control of various steels and non-ferrous alloys [6 Lectures]: Casting defects - fettling, inspection and testing of castings.

Text Books:
1. Serope Kalpakjian, (1997), Manufacturing processes for Engineering Materials, Addison Wesley.
2. Mukherjee, P. C. Fundamentals of metal casting technology, Oxford and IBH.

Reference Books:
1. Dieter G, Mechanical Metallurgy, Me Graw Hill, Kogakusha
2. Korolkove, V.Casting properties of metals and alloys

Self-Learning Material:
P K Jha, Principles of Metal Casting, NPTEL Course Material Department of Mechanical Engineering, IIT Roorkee https://nptel.ac.in/courses/112107215/

Title	**Welding Science and Technology**	Number	MEL4XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PE
Prerequisite	--		

Objectives:

The Instructor will:

1. Impart fundamental understanding of various conventional and non-conventional welding processes
2. Discuss various metallurgical aspects related with welding
3. Discuss heat transfer and associated issues with the welding operation

Learning Outcomes:

The students will have the ability to:

1. Appreciate the physics of welding operation and allied processes and select the process
2. Model heat transfer and cooling characteristics of welding

Contents:

Introduction and Classification of Welding [8 Lectures]: Introduction to welding processes and their applications; Welding as compared with other fabrication processes; classification of welding processes; design principles of welded structures; physics of welding arc; characteristics and stability of arc; modes of metal transfer; forces acting on a molten droplet; welding power sources and their characteristics.

Shielded Metal Arc Welding and Submerged Arc Welding [3 Lectures]: Principles, equipment, consumables, process parameters, and applications of Shielded Metal Arc Welding and Submerged Arc Welding.

Gas Tungsten Arc Welding and Gas Metal Arc Welding [4 Lectures]: Principles, equipment, shielding gases, process parameters, metal transfer characteristics, advantages, limitations, and applications of Gas Tungsten Arc Welding and Gas Metal Arc Welding.

Flux Cored Arc Welding, Plasma Arc Welding [2 Lectures]: Principles, process characteristics, advantages, and industrial applications of FCAW, and PAW processes.

Diffusion Welding [2 Lectures]: Principles, mechanisms, process variables, advantages, limitations, and applications of diffusion welding.

Friction Welding, Resistance Welding, Brazing, Soldering, and Surfacing [8 Lectures]: Principles, classifications, process parameters, equipment, advantages, limitations, defects, and industrial applications of friction welding, resistance welding processes, brazing, soldering, surfacing, and cladding techniques.

Fluxes and Coatings [2 Lectures]: Functions of welding fluxes and electrode coatings; types and classification of welding fluxes and coatings.

Heat Flow in Welding [4 Lecture]: Calculation of peak temperature; width of Heat Affected Zone; cooling rate and solidification rates; weld thermal cycles; residual stresses and their measurement; weld distortion and its prevention.

Testing of Welded Joints [4 Lectures]: Non-destructive testing (NDT) methods for welds; destructive testing methods for welded joints; weld defects and quality evaluation.

Welding Symbols, Standards, and Codes [2 Lectures]: Welding symbols for welding processes and positions; interpretation of welding symbols; relevant standards and codes used in welding practice.

Text/Reference books:

Text book:

1. Brien, O., (2004), Welding Handbook: Welding Processes, Part 1, Vol. 2, AWS.
2. Lancaster J. F., (1986), The Physics of welding, Pergamon.
3. Messler, R. W., (1999), Principles of Welding, John Wiley and Sons.

Reference book:

1. Grong O., (1997), Metallurgical modelling of welding, 2nd Ed, IOM Publication.
2. Radhakrishnan V M, (2002), Welding technology and design, New age.
3. Goldak A, (2005), Computational welding mechanics, Springer.
4. Lindgren L E, (2007), Computational welding mechanics, Woodhead Publishing Limited.

Self-Learning Material:

1. G Phanikumar, Analysis and Modeling of Welding, NPTEL Course Material Department of Metallurgical Engineering, IIT Madras, https://nptel.ac.in/courses/113106067/

Title	**Human Centered Automation and Industry 5.0**	Number	ME7XXX
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech	Type	PE
Prerequisite			

Objectives: The objectives of the course are to provide

1. Ability to understand and analyze human-centered, intelligent, and collaborative automation systems.
2. Awareness of sustainability, ethics, resilience, and safety in Industry 5.0 environments.

Learning Outcomes: The course will enable students to

1. Apply concepts of AI-assisted systems, digital twins, and adaptive automation in a human-centered industrial environment
2. Evaluate the sustainability, resilience, safety, and ethical aspects of modern automation systems.

Course Contents

Human-Centered Automation and Industry 5.0 Systems: Evolution from I4.0 to I5.0; Concept of human-centered automation, resilient industrial systems, collaborative industrial systems; Human augmentation versus replacement, human-machine-enterprise integration; Cognitive ergonomics, human reliability concepts, human workload and situational awareness models, adaptive automation systems. [7 lectures]

AI-Assisted Industrial Systems and Intelligent Decision Support: Human-in-the-loop AI systems; Explainable AI in industrial applications, Intelligent decision support systems, AI-assisted monitoring systems; Adaptive scheduling systems, Industrial copilots, data-driven operational support systems; Machine learning concepts for industrial applications, optimization-based industrial decision systems, intelligent enterprise systems. [8 lectures]

Zero Defect Manufacturing and Quality Analytics: Zero defect manufacturing concept; Statistical quality control, Process capability analysis, Six Sigma concepts, Intelligent quality control systems; In-line inspection systems, machine vision-based inspection, process-quality integration; closed-loop quality control, predictive quality systems, fault detection and defect prediction models, data-driven quality analytics. [8 lectures]

Intelligent Maintenance, Digital Twins, and Immersive Industrial Systems: Condition monitoring, predictive maintenance, and reliability analysis; Remaining useful life estimation; Human-centric digital twins and digital twin-driven monitoring systems; Synchronization of physical and virtual systems, augmented reality (AR) in maintenance and manufacturing, virtual reality (VR) for industrial training, mixed reality (MR) for operator assistance, immersive human-machine interaction systems. (10 Lectures)

Smart Enterprise Systems, Sustainability, and Responsible Automation: AI-assisted forecasting systems; Workforce-aware scheduling; Resilient supply chains and smart inventory systems; Blockchain-enabled traceability systems, sustainable automation systems, energy-aware industrial systems, lifecycle-aware automation; Ethical AI in industrial systems, trust in automation, worker privacy and data security, responsible and inclusive automation systems, multi-objective optimization concepts in industrial systems.

Reference Books

1. Longo F., Padovano A., and Umbrello S., *Human-Centric Smart Manufacturing*, Springer.
2. Tao F. and Qi Q., *Digital Twin Driven Smart Manufacturing*, Academic Press.
3. Nahavandi S., *Industry 5.0: A Human-Centric Solution*, Springer.

Title	**Vision for Automation**	Number	MEL7XXX
Department/IDRP	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for		Type	PE
Prerequisite	**RML6010/MEL7080**		

Objectives

1 Introduce machine vision concepts useful for robotics and automation applications
2 Introduce vision based precision 3D measurement systems used in robotics and automation

Learning Outcomes

1 Ability to design, calibrate and make measurements using machine vision systems and other equipment in robotics and automation applications

Contents:

Part 1: Machine Vision [13 Lectures]

Introduction to Machine Vision: Components, Applications [1 Lecture]

Hardware: Camera, Lens, Sensors - CCD, CMOS, Shutter types, Illumination models - Torrence sparrow model, Lambertian model, Transmission - USB, Ethernet, Camera-link, Image storage - types of image files jpg, bmp, etc., Design of a machine vision system, Define the environment and the task, Selection of lens, camera type, Selection of illumination type, Image processing, Measurements, Intelligence [3 Lectures]

Image Processing Techniques: Filtering, Convolution, Fourier transform, Thresholding, Morphology, Region Properties [4 Lectures]

Recognition: Textural Features for Engineering Surfaces, Case Studies [3 Lectures]

Deep learning-based Approaches: You Only Look Once, Generative Adversarial Networks [2 Lectures]

Part 2: 3D Vision for Robotics [13 Lectures]

3D Vision: Perspective Projection, Distortion, Triangulation

Camera Calibration: Direct Linear Transform, Tsai's method using Radial alignment Constraint, Zhang's method, Issues in Calibration, Perspective-n-Point [3 Lectures]

Multiple-view Geometry: Single View Metrology, Stereo Camera, Multiple View, Bundle adjustment [6 Lectures]

Sensor to Hand calibration: Hand-eye calibration, Extrinsic calibration of a 2D sensor [3 Lectures]

Machine Learning based Camera Calibration [1 Lecture]

Part 3: Precision Measurements for Robotics/Automation [13 Lectures]

Introduction to precision measurements for robotics/automation [1 Lecture]

Motion Capture Systems: Components, Principle, Measurement Process, Measurement for Robots, Demonstration, Measurement using 3D scanner: Components, Principle, Measurement Process [5 Lectures]

Robot calibration: Robot Performance measurements, base, tool, kinematic, elasto-static, Observability [5 Lectures]

Applications of pose measurement, 3D reconstruction, virtual reality, Machine Learning for 3D reconstruction and recognition [2 Lectures]

Reference Books

1 Jürgen Beyerer, Fernando Puente León, Christian Frese, Machine Vision: Automated Visual Inspection: Theory, Practice and Applications, Springer, 2016
2 Richard Szeliski, Computer Vision: Algorithms and Applications, Springer, 2022 (draft available at https://szeliski.org/Book/)
3 Zisserman and Hartley, Multiple View Geometry in Computer Vision, Cambridge University Press, 2004
4 Scientific articles:
  4.1 Z. Zhang, A flexible new technique for camera calibration, IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 22, No. 11, pp. 1330-1334, 2000
  4.2 Qilong Zhang and R. Pless, Extrinsic calibration of a camera and laser range finder (improves camera calibration), *2004 IROS,* vol.3, pp. 2301-2306, 2004
  4.3 Qi, Charles R., et al. Pointnet: Deep learning on point sets for 3d classification and segmentation., Proceedings of the IEEE conference on computer vision and pattern recognition, 2017.
  4.4 Qi, Charles Ruizhongtai, et al. "Pointnet++: Deep hierarchical feature learning on point sets in a metric space." Advances in neural information processing systems 30, 2017.
  4.5 Oleksandr Bogdan, Viktor Eckstein, Francois Rameau, Jean-Charles BazinAuthors Info & Claims, DeepCalib: a deep learning approach for automatic intrinsic calibration of wide field-of-view cameras, CVMP '18: Proceedings of the 15th ACM SIGGRAPH European Conference on Visual Media Production, Article No.: 6, Pages 1 – 10, 2018
5 ISO 9283:1998(en) Manipulating industrial robots-Performance criteria and related test methods

6 Hecht, E., Optics. Pearson Education India, 2012.

Self Learning Material

1 Berthold Horn. 6.801 Machine Vision. Fall 2020. Massachusetts Institute of Technology: MIT OpenCourseWare, https://ocw.mit.edu/courses/6-801-machine-vision-fall-2020/. License: Creative Commons BY-NC-SA. [https://www.youtube.com/playlist?list=PLUl4u3cNGP63pfpS1gV5P9tDxxL_e4W8O]

Title	**Combustion Technology**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	BTech, MTech and PhD	Type	ME
Prerequisite			

Objectives:

1. To introduce energy and combustion fundamentals for analyzing modern combustion systems.
2. To develop understanding of the basic principles and concepts of advanced fuel combustion and control process.
3. To be familiar with the fundamental physical and chemical principles regarding formation and control of air pollutants in industrial and technological processes.

Learning Outcomes:

1. Understanding of thermodynamics and kinetics of combustion.
2. Analysis of reacting systems and design of clean combustion systems

Contents:

1. Importance of combustion, Combustion phenomena in gas turbines, IC engines and power plants; Review of thermodynamics, Chemical kinetics, Mass transfer definitions: Fick's law. Equations of conservation of species mass, momentum, and energy; multicomponent diffusion equation, Schvab-Zel'dovich formulation, Rankine-Hugoniot relations **(13 lectures)**
2. Laminar premixed flames: Flame speed, adiabatic flame temperature, flammability limits, flame stabilization, ignition and quenching; Laminar diffusion flames: BurkeSchumann problem and droplet burning; Partially premixed flames **(13 lectures)**
3. Introduction to turbulent flames: premixed and diffusion flames; Detonations and Combustion instabilities **(6 lectures)**
4. Combustion of solids: solid propellant, drying, devolatilization and char combustion Spray combustion **(4 lectures)**
5. Chemical Emission from combustion, Quantification of emission, Emission control methods, Introduction to combustion modeling **(3 lectures)**

Text Book:

1. Turns, S. R. (2021), An Introduction to Combustion: Concepts and Applications, 4th Ed., McGraw-Hill

Reference Books:

1. Warnatz, J., Maas, U., & Dibble, R. W. (2006). Combustion: Physical and chemical fundamentals, modeling and simulation, experiments, pollutant formation (4th ed.). Springer.
2. Kuo K.K. (1986), Principles of Combustion, John Wiley and Sons

Online Material:

1. Fundamentals of Combustion, IIT Madras, Prof. V. Raghavan : https://nptel.ac.in/courses/112106299

Title	**Continuum Mechanics**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech.	Type	PE
Prerequisite	--		

Objective: The objective of the course is to

1. provide a rigorous theoretical foundation in nonlinear continuum mechanics for the modeling and analysis of deformable solids, developing a unified understanding of kinematics, stress, balance laws, and constitutive principles with strong emphasis on physical consistency through objectivity, material symmetry, and energy-based formulations.
2. bridge fundamental theory with practical engineering applications while preparing students for advanced study and research in solid mechanics, computational mechanics, fracture, biomechanics, and related fields.

Learning Outcomes: The students are expected to have the ability to:

1. Apply tensor and field-based tools to describe physical quantities and analyze deformation kinematics in material and spatial frameworks.
2. Formulate stress and traction measures, apply balance laws in spatial and referential forms, and analyze stress using invariants.
3. Develop objective, symmetry-consistent constitutive models and analyze finite deformation and nonlinear elastic behavior through energy-based approaches.

Contents:

Mathematical Foundations [6 Lectures]*:* Scalars, vectors, and tensors; index notation; tensor products and contractions; invariants; eigenvalues and eigenvectors of symmetric tensors; change of basis and transformation laws; scalar, vector, and tensor fields; differential operators.

Kinematics [8 Lectures]: Bodies and configurations; reference and current descriptions; motion, displacement, velocity, and acceleration fields; material and spatial derivatives; deformation gradient; transformation of line, area, and volume elements; polar decomposition; strain measures and linearized kinematics; velocity gradient, rate of deformation, and spin tensor; rigid body motions and objectivity in kinematics.

Stress Measures and the Concept of Stress [6 Lectures]: Traction vectors and Cauchy's postulate; existence and interpretation of the Cauchy stress tensor; stress transformation, principal and extremal stresses, and stress invariants; alternative stress measures including the first and second Piola-Kirchhoff stresses; physical interpretation through representative stress states.

Mechanical Balance Laws and Field Equations [5 Lectures]: Conservation of mass; Reynolds transport theorem; balance of linear and angular momentum; balance of mechanical energy.

Constitutive Principles and Objectivity [5 Lectures]: Constitutive response functions; material frame indifference and change of observer; objective scalar, vector, and tensor fields; objective stress rates; material symmetry and isotropy; linearization of constitutive laws.

Nonlinear Elasticity and Hyperelastic Materials [5 Lectures]: Hyperelastic material concept; strain energy density functions; compressible and incompressible responses; classical hyperelastic models including Neo-Hookean, Mooney Rivlin, Ogden, Gent, etc.

Applications [4 Lectures]: Homogeneous deformation of hyperelastic membranes under uniaxial and biaxial loading, and simple shear; balloon inflation, instabilities of soft materials: snap through instability and morphological instabilities: wrinkles, creases, folds, and ridges.

Text books

1. G.A. Holzapfel, Nonlinear Solid Mechanics - A continuum approach for engineering, 2000
2. Jog, C.S., 2015. Continuum mechanics (Vol. 1). Cambridge University Press.

Reference books

1. Volokh, K., 2016. Mechanics of soft materials (Vol. 337). Singapore: Springer.
2. Haupt, P., 2013. Continuum mechanics and theory of materials. Springer Science & Business Media.
3. Truesdell, C. and Noll, W., 2004. The non-linear field theories of mechanics. In The non-linear field theories of mechanics (pp. 1-579). Berlin, Heidelberg: Springer Berlin Heidelberg.

4. Abeyaratne, R., 1998. Continuum mechanics. Lecture Notes on The Mechanics of Elastic Solids.

Online course material

1. Continuum Mechanics, NPTEL, https://nptel.ac.in/courses/112103167
2. Lectures on Continuum Physics, Open Umich, https://open.umich.edu/find/open-educational-resources/engineering/lectures-continuum-physics

Title	**Engineering Fracture Mechanics**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech./M.Tech.	Type	PE
Prerequisite	Mechanics of Solids		

Objective: The objective of the course is to

1. Provide the fundamentals of linear elastic fracture mechanics,
2. Introduce various numerical approaches for solving fracture problems.

Learning Outcomes: After completing this course, students will be able to

1. Solve fracture problems numerically using both discrete and diffuse approaches.
2. Interpret numerical results.

Contents

Background [3 Lectures]: Historical development, Review of crack-tip fields in linear elastic fracture mechanics (LEFM); Modes of fracture (I, II, III), Analytical solution of plane crack problems, stress intensity factors, energy release rate, and their equivalence.

Energy release rate [10 Lectures]: Griffith theory and extensions (Irwin, Orowan); Critical energy release rate; R-curve behavior and fracture stability, concept of J-integral (path independence), Relation between G and J, Introduction to elastic-plastic fracture mechanics (EPFM) parameters (CTOD, J as nonlinear energy release rate), calculation of SIF (analytical, experimental, numerical methods).

Review of Finite Element Method [5 Lectures]: Basic concepts of FEM, weak formulation of boundary value problems, discretization, interpolation functions, FE-formulation for scalar and vector valued problems, Assembly of global equations, and solution procedure.

Discrete approach for fracture simulation [13 Lectures]: FE for the cracked body, Need for mesh refinement near the crack tip, Singular and quarter-point elements, Method for determining fracture parameters, Domain and interaction integral approach. Extended finite element method, enrichment functions: crack tip and heavyside functions, Cohesive zone model.

Diffuse approach for fracture simulation [13 Lectures]: Introduction of various diffuse approaches, Phase-field model for fracture, derivation of governing equations from variational principle, Length-scale parameter and regularization; Phase-field for brittle fracture (AT model, phase-field evolution); Extensions to fatigue; Advantages over discrete methods (no explicit crack tracking, natural branching/coalescence); Numerical implementation (staggered/monolithic solvers).

Test Books:

1. Kumar, P., 2009. Elements of fracture mechanics. McGraw-Hill Education LLC.
2. Anderson, T.L. and Anderson, T.L., 2005. Fracture mechanics: fundamentals and applications. CRC press.

Reference Books:

1. Broek, D., 1982. Elementary engineering fracture mechanics. Springer Science & Business Media.
2. Kuna, M., 2013. Finite elements in fracture mechanics (Vol. 201). Dordrecht: Springer Netherlands.
3. Bordas, S.P., Menk, A. and Natarajan, S., 2023. Partition of unity methods. John Wiley & Sons

Self-learning:

1. Engineering Fracture Mechanics, NPTEL, https://nptel.ac.in/courses/112106065

Title	**Mechanics of Smart Materials and Structures**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	M.Tech. and B.Tech.	Type	PE
Prerequisite	--		

Objective: The objective is to

1. provides a rigorous theoretical foundation in the mechanics of smart materials and structures, focusing on deformable systems governed by coupled mechanical, electrical, and magnetic fields.
2. Provide understanding of multiphysics coupling, structural response, and energy-based constitutive principles grounded in thermodynamic consistency, work conjugacy, reciprocity, and stability.

Learning Outcomes: The students are expected to have the ability to:

1. Apply continuum and field-based mechanics concepts to describe coupled mechanical, electrical, and magnetic phenomena in deformable materials and structural systems.
2. Formulate and analyze energy-based constitutive relations for electromechanical and magnetomechanical coupling, ensuring physical consistency through work conjugacy, reciprocity, and stability.
3. Model and assess smart structures incorporating active materials, architected geometries, and periodic media for sensing, actuation, tunability, and wave control.

Contents:

Introduction [2 lectures]: Introduction to smart materials and smart structures, Classification of smart materials, Applications in sensing, actuation, and energy harvesting.

Review of Non-linear Mechanics [7 Lectures]: Kinematics: Configurations and motion of continuum bodies, deformation gradient, strain tensors; Stress and the balance principles: conservation of mass, the momentum principles, the Cauchy stress tensor, stress measures for large deformations, the equations of motion and symmetry of stress, balance of mechanical energy, the principles of virtual work and power.

Piezoelectric Materials [6 Lectures]: History of Piezoelectricity, Piezoelectric Materials, Direct and converse piezoelectric effects; constitutive behaviour and governing equations; electromechanical coupling coefficients and energy conversion, modeling and engineering applications.

Electroactive Polymers and Soft Actuation [6 Lectures]: Electro-active polymers and their classification, electrostatic actuation mechanisms, Dielectric elastomers, nonlinear electromechanical coupling and governing equations, large deformation response; electro-mechanical instabilities and pull-in phenomena; modeling and engineering applications.

Magneto-Active Materials [6 Lectures]: Magnetostriction; hard and soft magneto-active elastomers, magnetorheological elastomers; magneto-mechanical constitutive behavior; magnetic body forces and Maxwell stress; magneto-mechanical instabilities, modeling and engineering applications.

Architected Materials and Structures [6 Lectures]: Homogenization and effective properties of heterogeneous media; lattice and cellular structures; auxetic response and negative Poisson's ratio; architected stiffness and damping; structural instabilities as design tools; metastructures with embedded smart materials for tunable response.

Wave Control and Tunable Phononic Structures [6 Lectures]: Wave propagation in periodic media; dispersion relations; Bragg scattering and bandgap formation; locally resonant structural systems; vibration attenuation and isolation; active and tunable phononic structures using smart materials.

Text books

1. Gandhi, M.V. and Thompson, B.D., 1992. Smart materials and structures. Springer Science & Business Media.
2. Gaudenzi, P., 2009. Smart structures: physical behaviour, mathematical modelling and applications. John Wiley & Sons.

Reference books

1. Cady, W.G., 2018. Piezoelectricity: volume two: an introduction to the theory and applications of electromechanical phenomena in crystals. Courier Dover Publications.
2. Dorfmann, L. and Ogden, R.W., 2014. Nonlinear theory of electroelastic and magnetoelastic interactions (Vol. 1). New York: Springer.
3. Holzapfel, G.A., 2002. Nonlinear solid mechanics: a continuum approach for engineering science.
4. Volokh, K., 2016. Mechanics of soft materials (Vol. 337). Singapore: Springer.
5. Deymier, P.A. ed., 2013. Acoustic metamaterials and phononic crystals (Vol. 173). Springer science & Business media.
6. Lakes, R., 2020. Composites and metamaterials. World Scientific.

Online course material

1. Smart Materials and Intelligent System Design, NPTEL, nptel.ac.in/courses/112104251

Title	**Structural Health Monitoring**	Number	MEL7XX0
Department	Mechanical Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech./M.Tech.	Type	Elective
Prerequisite	Solid Mechanics, Mechanics of Composite Materials		

Objectives:

1. The objective of this course is to provide students with a comprehensive understanding of the principles, technologies, and methodologies of Structural Health Monitoring (SHM) for engineering structures.
2. The course aims to develop a strong foundation in SHM philosophy, sensing mechanisms, data acquisition, and decision-making frameworks, with an emphasis on smart materials, including piezoelectric, magnetostrictive, and optical fiber sensors.

Learning Outcomes: The student will have ability to

1. Analyze local vibration and impedance-based SHM data using piezoelectric and smart material-based sensing techniques.
2. Select appropriate sensors, install SHM systems on real structures, and evaluate sensor performance and diagnostics.
3. Apply global vibration techniques to extract modal parameters such as natural frequencies and mode shapes for damage detection.

Contents-

Introduction to SHM [10 Lectures]: An overview of Structural Health Monitoring: definition, objectives, and philosophy. SHM framework: sensing, data acquisition, feature extraction, and decision making. SHM versus Non-Destructive Evaluation (NDE): similarities and differences. Levels of damage identification: detection, localization, quantification, prognosis. Overview of sensor systems and performance requirements for SHM. Smart materials for SHM: definition and classification. Piezoelectric materials: basic physics, constitutive relations, sensing and actuation. Magnetostrictive materials: fundamentals, magnetization and hysteresis behaviour. Optical fiber sensors and Lamb wave–based SHM methods. Emerging SHM technologies using piezo, magnetostrictive, and optical fiber sensors. Solution domain for SHM and commonly used damage indices. Overview of application potential of SHM in mechanical, aerospace, civil infrastructures, and other engineering systems.

Sensing and Data Acquisition System [10 Lectures]: Strain gauges, displacement sensors, accelerometers. Fiber Bragg Grating (FBG) sensors and optical sensing principles. Acoustic emission sensors and piezoelectric transducers. Embedded and surface-mounted sensors for composites, metals, and concrete. Wireless sensor networks, energy harvesting sensors, and sensor placement optimization. Data acquisition hardware/software and sampling requirements.

Structural Health Monitoring with Piezoelectric and Magnetostrictive Sensors [11 Lectures]: Sensor technology, Available industrial acousto-ultrasonic systems with piezoelectric sensors, Electro-Mechanical Impedance (EMI) technique, Delamination Sensing using Piezo Sensory Layer. Voltage response from Piezopatch. SHM using Magnetostrictive sensory layer. Basics of Magnetization and Hysteresis. Delamination sensing using Magnetostrictive sensory layer. Constitutive equation for the composite. Magnetostrictive sensory layer in symmetric laminate.

Structural Health Monitoring using Laser Doppler Velocimeter (LDV) or Vibrometer [8 Lectures]: Introduction to LDV. Experimental Modal Analysis using LDV. Linking experimental and analytical vibration data. Modal Assurance Criterion (MAC) for mode pairing. Velocity and Displacement Measurement using LDV. Mathematical description of structural system with damage. Damage localization and quantification. Case study for Symmetric and Cross-ply laminates.

Text Books:

1. Soh, C.K., Yang, Y. and Bhalla, S. eds., 2012. Smart materials in structural health monitoring, control and biomechanics. Springer Science & Business Media.

2. Bhalla, S., Moharana, S., Talakokula, V. and Kaur, N., 2017. Piezoelectric materials: applications in SHM, energy harvesting and biomechanics.

Reference Books:

1. Balageas, D., Fritzen, C.P. and Güemes, A. eds., 2010. Structural health monitoring (Vol. 90). John Wiley & Sons.
2. Gandhi, M.V. and Thompson, B.D., 1992. Smart materials and structures. Springer Science & Business Media.
3. Ewins, D.J., 2009. Modal testing: theory, practice and application. John Wiley & Sons.

Self-Learning Materials

1. Structural Health Monitoring, NPTE, https://nptel.ac.in/courses/114106046